

Contents

MATH 265 Exam Prep Guide	1
How to read the two notations in this file	1
What I analyzed	1
Exam facts that matter	1
Möbius answer-entry rules	2
Part 1: The big picture and limits	2
1. Calculus is all about performing two operations on functions	2
2. Rate of change as slope of a straight line	2
3. The dilemma of the slope of a curvy line	3
4. The slope between very close points	3
5. The limit	3
6. The derivative, and differentials of x and y	4
7. Differential notation	4
Part 2: Derivative rules	4
8. The constant rule of differentiation	4
9. The power rule of differentiation	5
10. Visual interpretation of the power rule	5
11. The addition and subtraction rule of differentiation	5
12. The product rule of differentiation	6
13. Combining rules of differentiation to find the derivative of a polynomial	6
14. Differentiation super-shortcuts for polynomials	6
15. Solving optimization problems with derivatives	7
16. The second derivative	7
17. Trig rules of differentiation for sine and cosine	7
18. Knowledge test: product rule example	8
19. The chain rule for differentiation	8
20. The quotient rule for differentiation	8
21. The derivative of the other trig functions	9
22. Algebra overview: exponentials and logarithms	9
23. Differentiation rules for exponents	10
24. Differentiation rules for logarithms	10
Part 3: Antiderivatives and integration	11
25. The anti-derivative, also called the integral	11
26. The power rule for integration	11
27. The power rule for integration will not work for $1/x$	11
28. The constant of integration, plus C	12
29. Anti-derivative notation	12
30. The integral as the area under a curve using the limit	12
31. Evaluating definite integrals	13
32. Definite and indefinite integrals compared	13
33. The definite integral and signed area	13
34. The Fundamental Theorem of Calculus visualized	14
35. The integral as a running total of its derivative	14
36. The trig rule for integration, sine and cosine	15
37. Definite integral example problem	15
38. u -Substitution	15
39. Integration by parts	16
40. The DI method for integration by parts	16
Part 4: Other must-know topics from your exams	17
Functions, domains, and composition	17
Exact trig values	17

Table 1 of 2 - sine, cosine and tangent	18
Table 2 of 2 - cosecant, secant and cotangent	19
Graph transformations	20
Limits and continuity	21
Asymptotes	21
Tangent lines and perpendicular slopes	21
Implicit differentiation	22
Related rates	22
Linear approximation and differentials	22
Newton's method	23
Curve sketching with calculus	23
Extreme values on a closed interval	23
Optimization	23
Properties of the definite integral and bounding	24
Area between curves	24
Net change and motion	24
Work	24
Average value of a function	25

Part 5: Every exam question, restated and solved **25**

1. Functions and precalculus foundations	25
Q1.1 - Exact value of a cosine	25
Q1.2 - Composite functions and their domains	26
Q1.3 - Which equations define a function of x	27
Q1.4 - Volume of a cone as a function of its radius	28
2. Graph transformations	29
Q2.1 - Graph a shifted parabola by transformations	29
Q2.2 - Graph a reflected absolute value by transformations	30
3. Limits and continuity	30
Q3.1 - Five limits, algebraic and trigonometric	30
Q3.2 - Six limits, with justification required	32
Q3.3 - Two limits, at infinity and at an asymptote	35
4. Asymptotes	36
Q4.1 - Vertical and horizontal asymptotes of a rational function	36
5. Graphical interpretation and sketching from conditions	37
Q5.1 - Sketch a function from four conditions	37
Q5.2 - Interpret five conditions, then sketch	37
Q5.3 - Interpret six conditions, then sketch	38
6. Rates of change - interpretation	39
Q6.1 - Average and instantaneous rate of population change	39
7. The derivative from its definition	39
Q7.1 - Derivative of cotangent from the definition	39
8. Derivative rules and computation	41
Q8.1 - Four derivatives, power chain and quotient rules	41
Q8.2 - Four derivatives, including a second derivative	42
Q8.3 - Five derivatives, nested chain and quotient rules	44
Q8.4 - Two derivatives, naming the rules used	46
9. Tangent lines	47
Q9.1 - Tangent line perpendicular to a given line	47
10. Implicit differentiation	48
Q10.1 - Find the derivative implicitly	48
Q10.2 - Tangent line to an implicit curve	49
11. Related rates	50
Q11.1 - Gravel cone growing in height	50
Q11.2 - Rocket tracked by a radar station	51
12. Linear approximation and differentials	52
Q12.1 - Approximate a square root with differentials	52
Q12.2 - Estimate a sine with linearization	52

Q12.3 - Two estimates to four decimal places	53
13. Newton's method	54
Q13.1 - Approximate a root with Newton's method	54
14. Curve sketching with calculus	55
Q14.1 - Sketch two curves with a full calculus analysis	55
15. Extreme values and optimization	57
Q15.1 - All maxima and minima on closed intervals	57
Q15.2 - Rectangle of least perimeter	59
Q15.3 - Absolute extremes by the Extreme Value Theorem	60
Q15.4 - Cheapest rectangular box	61
Q15.5 - Absolute extremes of a function with a cusp	62
16. Antiderivatives and integration techniques	63
Q16.1 - Five integrals	63
Q16.2 - Four integrals, naming the technique	66
17. Properties of the definite integral	68
Q17.1 - Interval containing a secant integral	68
Q17.2 - Estimate an integral from its properties	69
Q17.3 - Bound an integral from its properties	70
18. The Fundamental Theorem of Calculus	70
Q18.1 - Differentiate an integral with variable limits	70
19. Applications of integration	71
Q19.1 - Area between a line and a parabola	71
Q19.2 - Area between a cosine and a horizontal line	72
Q19.3 - Net change, water flowing from a tank	74
Q19.4 - Work, raising one end of a chain	74
Q19.5 - Work, stretching a spring	75
Q19.6 - Average temperature of a metal rod	76
Q19.7 - Find k from an average value	76
Q19.8 - Displacement and distance travelled	77
Q19.9 - Sprinter with piecewise acceleration	79
Part 6: The one-page formula card	79
Part 7: Practice map	82
Part 8: Weekly practice plan	83
Part 9: Final checklist before exam day	84
The ten most expensive single mistakes in this course	84

MATH 265 Exam Prep Guide

This file is a phone-friendly study guide for the MATH 265 midterm and final. It covers the full concept list you gave, plus the extra topics your own notes and sample exams actually test.

How to read the two notations in this file

Every formula in this guide is given twice, on purpose.

- The **backtick line** is the Möbius / calculator form: the flat, one-line text you actually type into an answer box.
- The **Math line** under it is the same formula in ordinary mathematical symbols, the way it is printed in the textbook and the way you should write it on a paper exam.

Example of the pairing you will see everywhere below:

- $(f/g)' = (f' g - f g') / g^2$.

- Math: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

Read the Math line to understand the formula. Read the backtick line when you are about to type it. The two always say the same thing.

What I analyzed

- The course folder at Coursework_Archive\Degree\Athabasca_MATH265 - Calculus 1.
- The question notes in notes\Q01.1 through notes\Q19.9.
- The review and cheat-sheet files in notes\MATH265.
- The four sample exams, the all-exam question bank, and the AU reference sheets.
- The question bank notes\MATH265.md, which supplies the exact wording of every question re-stated in Part 5.

Key findings from the files:

- The midterm covers Units 1 to 4.
- The final covers Units 3 to 7.
- Both exams are 3 hours, closed-book, and need 50 percent to pass.
- You may bring one 8.5 by 11 inch note sheet, both sides, and a simple scientific calculator.
- The sample papers split cleanly: midterms stop at differentials and related rates.
- Finals add integration, curve sketching, extrema, optimization, Newton's method, and applications of integration.
- The reference sheet notes that the published sample exams use no integration by parts, but you asked for it, so it is included below as insurance.

Exam facts that matter

- Midterm is 25 percent of the final grade.
- Final is 40 percent of the final grade.
- Assignments are the rest.
- You must pass each exam on its own, so do not rely on assignments to rescue you.
- Questions are worth several points because several steps are required.
- Show every step and justify every answer.
- Use radians unless the question says degrees.
- Use exact form unless the question asks for rounding.
- Do the questions you are confident about first.
- Budget roughly 10 to 15 minutes per question.

Möbius answer-entry rules

The live delivery is machine-marked, and it grades by mathematical equivalence. A correct answer is never rejected for its form, only for being parsed differently than you meant it.

- Multiplication is explicit: write $2*x$, never $2x$.
 - Roots are functions: write $\text{sqrt}(6)$, never a root glyph and never $6^{\cdot}5$.
 - Bracket whole numerators: write $(\text{sqrt}(6)+\text{sqrt}(2))/4$, never $\text{sqrt}(6)+\text{sqrt}(2)/4$.
 - Natural log is $\ln(x)$, never $\log(x)$.
 - Exact form beats decimals: write $(\text{sqrt}(6)+\text{sqrt}(2))/4$, never 0.9659 .
 - Powers use a caret: write $x^{(3/2)}$, and bracket any compound exponent.
 - Always click the preview icon before you submit.
 - An empty preview modal means broken syntax, not wrong mathematics.
-

Part 1: The big picture and limits

1. Calculus is all about performing two operations on functions

Concept:

- Calculus has two main operations: differentiation and integration.
- Differentiation finds how fast a function changes.
- Integration adds up how much a function accumulates.
- The two operations are inverses, which is the Fundamental Theorem of Calculus.

Know:

- Derivative = rate of change = slope.
- Integral = accumulation = area under a curve.
- Most exam problems are one of these two operations applied to a function.

Practice:

- Say, in one sentence, whether a word problem wants a derivative or an integral.
- Example: water flowing from a tank asks for total amount, so integrate the rate.
- Example: population growing at an instant asks for a derivative.

2. Rate of change as slope of a straight line

Concept:

- A straight line has one slope, the same everywhere.
- Slope = rise over run = change in y divided by change in x.

Know:

- Average rate of change on an interval is the slope of the secant line.
- For a line through two points, the slope is $(y_2 - y_1) / (x_2 - x_1)$.
 - Math: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$

Practice:

- Interpret $(P(15) - P(2)) / 13 = 1,431$ in plain words.
 - Math: $\frac{P(15) - P(2)}{13} = 1431$
- Say the population grew by an average of 1,431 people per year between year 2 and year 15.
- See Q06.1.

3. The dilemma of the slope of a curvy line

Concept:

- A curve does not have one slope because its steepness changes point to point.
- A straight secant between two far points only gives an average.

Know:

- To get steepness at a single point, you have to zoom in.
- The slope at one point is the slope of the tangent line there.

Practice:

- Draw a parabola and mark a secant between two points.
- Then draw the tangent line at one of those points.
- Explain why the secant slope is not the same as the tangent slope.

4. The slope between very close points

Concept:

- Move the two points closer and closer together.
- The secant slope gets closer to the tangent slope.

Know:

- The difference quotient is $(f(x + h) - f(x)) / h$.
 - Math: $\frac{f(x + h) - f(x)}{h}$
- As h approaches 0, this quotient approaches the derivative.
 - Math: $\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = f'(x)$

Practice:

- Write the difference quotient for $f(x) = x^2$.
 - Math: $f(x) = x^2$
- Simplify it to $2x + h$, then let h go to 0 to get $2x$.
 - Math: $\frac{(x + h)^2 - x^2}{h} = 2x + h \rightarrow 2x$

5. The limit

Concept:

- A limit is the value a function approaches as the input approaches a value.
- The function does not have to actually reach that value.

Know:

- Direct substitution is the first thing to try.
- $0/0$ means factor, rationalize, use a common denominator, or use a trig identity.
 - Math: $\frac{0}{0}$ is an indeterminate form, so the expression must be rewritten before the limit is read off.
- $k/0$ with k not zero usually means the limit is infinite or does not exist.
 - Math: $\frac{k}{0}$ with $k \neq 0$ gives $\pm\infty$, so check the sign from each side.
- infinity/infinity means divide by the highest power in the denominator.
 - Math: $\frac{\infty}{\infty}$ is indeterminate, so divide numerator and denominator by the highest power of x in the denominator.
- Bounded times zero, such as (something going to 0) * $\sin(x)$, is squeezed to 0.
 - Math: if $\lim_{x \rightarrow a} g(x) = 0$ and $|b(x)| \leq M$, then $\lim_{x \rightarrow a} g(x)b(x) = 0$.

Practice:

- Do every limit in Q03.1, Q03.2, and Q03.3.
- Also do Q04.1 to connect limits to asymptotes.

6. The derivative, and differentials of x and y

Concept:

- The derivative is the limit of the difference quotient.
- It measures instantaneous rate of change.

Know:

- $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)] / h$.

- Math: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

- The derivative is a function, not a single number.
- dy/dx means the derivative of y with respect to x .

- Math: $\frac{dy}{dx}$

Practice:

- Use the definition to find the derivative of x^2 , $1/x$, and $\cot x$.
 - Math: x^2 , $\frac{1}{x}$, $\cot x$
- See Q07.1 for the cotangent derivation.

7. Differential notation

Concept:

- dy/dx is one notation for the derivative.
- dy and dx are called differentials.

Know:

- $dy = f'(x) dx$.
 - Math: $dy = f'(x) dx$
- Treat dx as a small change in x .
- Treat dy as the approximate resulting change in y .
- Differential notation is used in linear approximation and related rates.

Practice:

- Approximate $\sqrt{9.2}$ by choosing $x = 9$ and $dx = 0.2$.
 - Math: $\sqrt{9.2}$ with $a = 9$ and $dx = 0.2$
 - See Q12.1.
-

Part 2: Derivative rules

8. The constant rule of differentiation

Concept:

- The derivative of a constant is zero.

Know:

- $d/dx [c] = 0$.
 - Math: $\frac{d}{dx}(c) = 0$
- A constant function has a flat graph, so its slope is zero.

Practice:

- Find $d/dx [7]$.
 - Math: $\frac{d}{dx}(7)$
- Answer: 0.
 - Math: 0

9. The power rule of differentiation

Concept:

- Bring the exponent down and reduce the exponent by one.

Know:

- $d/dx [x^n] = n x^{(n-1)}$.
 - Math: $\frac{d}{dx} (x^n) = n x^{n-1}$
- Works for negative and fractional powers too.

Practice:

- Differentiate x^5 , x^{-2} , $\sqrt{x}$, and $1/x$.
 - Math: x^5 , x^{-2} , $\sqrt{x}$, $\frac{1}{x}$
- Answers: $5x^4$, $-2x^{-3}$, $(1/2)x^{-1/2}$, $-x^{-2}$.
 - Math: $5x^4$, $-2x^{-3}$, $\frac{1}{2}x^{-1/2}$, $-x^{-2}$

10. Visual interpretation of the power rule

Concept:

- Each derivative lowers the degree of a polynomial by one.
- The graph of the derivative has one fewer turn than the original.

Know:

- A cubic has a quadratic derivative.
- A quadratic has a linear derivative.
- A line has a constant derivative.

Practice:

- Sketch $y = x^3$ and $y = 3x^2$ together.
 - Math: $y = x^3$ and $y = 3x^2$
- Notice the cubic's flat points correspond to where $3x^2 = 0$.
 - Math: $3x^2 = 0$

11. The addition and subtraction rule of differentiation

Concept:

- Differentiate term by term.

Know:

- $d/dx [f(x) + g(x)] = f'(x) + g'(x)$.
 - Math: $\frac{d}{dx} (f(x) + g(x)) = f'(x) + g'(x)$
- $d/dx [f(x) - g(x)] = f'(x) - g'(x)$.
 - Math: $\frac{d}{dx} (f(x) - g(x)) = f'(x) - g'(x)$

Practice:

- Differentiate $4x^5 + 3x^4 - 6x^3 + 6$.
 - Math: $4x^5 + 3x^4 - 6x^3 + 6$
- Answer: $20x^4 + 12x^3 - 18x^2$.
 - Math: $20x^4 + 12x^3 - 18x^2$

12. The product rule of differentiation

Concept:

- For a product of two functions, use first derivative times second plus first times second derivative.

Know:

- $(fg)' = f'g + fg'$.
- Math: $(fg)' = f'g + fg'$
- Say it: "first prime times second, plus first times second prime."

Practice:

- Differentiate $x \cos(\sqrt{x-3})$.
- Math: $x \cos(\sqrt{x-3})$
- Identify $f = x$ and $g = \cos(\sqrt{x-3})$.
- Math: $f = x$ and $g = \cos(\sqrt{x-3})$

13. Combining rules of differentiation to find the derivative of a polynomial

Concept:

- Polynomials use only the power, constant, and sum rules.
- More complicated functions layer product, quotient, and chain rules.

Know:

- Work from the outside in.
- Name the rule you are using at each step.

Practice:

- Differentiate a polynomial like $3x^4 - 2x^3 + x - 5$.
- Math: $3x^4 - 2x^3 + x - 5$
- Then move on to mixed expressions in Q08.1 through Q08.4.

14. Differentiation super-shortcuts for polynomials

Concept:

- For a polynomial, multiply each coefficient by its exponent and drop the degree by one.

Know:

- $y = ax^n$ becomes $y' = a n x^{(n-1)}$.
- Math: $y = ax^n$ becomes $y' = an x^{n-1}$
- The constant term disappears.

Practice:

- Differentiate $y = 4x^5 + 3x^4 - 6x^3 + 6$ without rewriting every step.
- Math: $y = 4x^5 + 3x^4 - 6x^3 + 6$
- Answer: $20x^4 + 12x^3 - 18x^2$.
- Math: $20x^4 + 12x^3 - 18x^2$

15. Solving optimization problems with derivatives

Concept:

- Maxima and minima happen where the derivative is zero or undefined, or at endpoints.

Know:

- Draw and label the situation.

- Write the quantity to optimize.
- Write the constraint.
- Use the constraint to eliminate one variable.
- Differentiate, set the derivative to zero, and solve.
 - Math: solve $f'(x) = 0$
- Check endpoints and state the answer with units.

Practice:

- Rectangle of fixed area and least perimeter.
- Cheapest rectangular box with different lid cost.
- See Q15.2, Q15.4, and Q15.1.

16. The second derivative

Concept:

- The second derivative is the derivative of the derivative.

Know:

- $f'(x)$ tells slope and velocity.
 - Math: $f'(x)$
- $f''(x)$ tells concavity and acceleration.
 - Math: $f''(x) = \frac{d^2f}{dx^2}$
- $f'' > 0$ means concave up.
 - Math: $f''(x) > 0$
- $f'' < 0$ means concave down.
 - Math: $f''(x) < 0$

Practice:

- Find $d^2/dx^2 [\cot(2x)]$.
 - Math: $\frac{d^2}{dx^2} \cot(2x)$
- See Q08.2.

17. Trig rules of differentiation for sine and cosine

Concept:

- The derivative of sine is cosine.
- The derivative of cosine is negative sine.

Know:

- $d/dx [\sin x] = \cos x$.
 - Math: $\frac{d}{dx}(\sin x) = \cos x$
- $d/dx [\cos x] = -\sin x$.
 - Math: $\frac{d}{dx}(\cos x) = -\sin x$
- The minus sign on cosine is the most common lost mark.

Practice:

- Differentiate $\sin(2x^2 - x + 1)$.
 - Math: $\sin(2x^2 - x + 1)$
- Remember the chain rule: multiply by the inside derivative $4x - 1$.
 - Math: multiply by $4x - 1$

18. Knowledge test: product rule example

Concept:

- This is a self-test to confirm you can apply the product rule correctly.

Know:

- For $y = x^2 \sin x$, let $f = x^2$ and $g = \sin x$.
 - Math: for $y = x^2 \sin x$, let $f = x^2$ and $g = \sin x$
- $f' = 2x$ and $g' = \cos x$.
 - Math: $f' = 2x$ and $g' = \cos x$
- $y' = 2x \sin x + x^2 \cos x$.
 - Math: $y' = 2x \sin x + x^2 \cos x$

Practice:

- Differentiate $x^2 \sin x$ on paper.
 - Math: $x^2 \sin x$
- Then check your answer against $2x \sin x + x^2 \cos x$.
 - Math: $2x \sin x + x^2 \cos x$

19. The chain rule for differentiation

Concept:

- Differentiate the outside function, then multiply by the derivative of the inside.

Know:

- $d/dx [f(g(x))] = f'(g(x)) * g'(x)$.
 - Math: $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$
- Work outside to inside.
- Missing the inside derivative is the single most common error.

Practice:

- Differentiate $\sin(2x^2 - x + 1)$.
 - Math: $\sin(2x^2 - x + 1)$
- Differentiate $(-4x^3 - x^2 + 3x + 7)^4$.
 - Math: $(-4x^3 - x^2 + 3x + 7)^4$
- See Q08.1 and Q08.3.

20. The quotient rule for differentiation

Concept:

- For a fraction, use low d-high minus high d-low over low squared.

Know:

- $(f/g)' = (f'g - fg') / g^2$.
 - Math: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- The numerator order is the whole mark.
- Never write $g f' - f g'$.
 - Math: never write $\frac{gf' - fg'}{g^2}$ with the terms swapped, and never drop the square on the denominator.

Practice:

- Differentiate $(2x - 16) / (x + 3)^2$.
- Math: $\frac{2x - 16}{(x + 3)^2}$
- Differentiate $(\sqrt{x^2 - 1}) / (x^2 - 2x - 8)$.
- Math: $\frac{\sqrt{x^2 - 1}}{x^2 - 2x - 8}$

21. The derivative of the other trig functions

Concept:

- Tangent, cotangent, secant, and cosecant each have their own derivative.

Know:

- $d/dx [\tan x] = \sec^2 x$.
- Math: $\frac{d}{dx}(\tan x) = \sec^2 x$
- $d/dx [\cot x] = -\csc^2 x$.
- Math: $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $d/dx [\sec x] = \sec x \tan x$.
- Math: $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $d/dx [\csc x] = -\csc x \cot x$.
- Math: $\frac{d}{dx}(\csc x) = -\csc x \cot x$
- Every co-function derivative has a minus sign: cosine, cotangent, cosecant.

Practice:

- Differentiate $\sec(x^2 - 3x)$.
- Math: $\sec(x^2 - 3x)$
- Differentiate $\cot(2x)$ twice.
- Math: $\cot(2x)$
- See Q08.2.

22. Algebra overview: exponentials and logarithms

Concept:

- Exponentials grow by repeated multiplication.
- Logarithms undo exponentials.

Know:

- e^x is the natural exponential.
- Math: e^x
- $\ln x$ is the natural logarithm, the inverse of e^x .
- Math: $\ln x$ is the inverse of e^x
- $\ln(e^x) = x$ and $e^{\ln x} = x$.
- Math: $\ln(e^x) = x$ and $e^{\ln x} = x$
- Exponent rules: $x^a x^b = x^{(a+b)}$, $1/x^n = x^{(-n)}$, and $\text{nth-root}(x^m) = x^{(m/n)}$.
- Math: $x^a x^b = x^{a+b}$, $\frac{1}{x^n} = x^{-n}$, and $\sqrt[n]{x^m} = x^{m/n}$

Practice:

- Rewrite $\sqrt{3x}$ as $\sqrt{3} x^{(1/2)}$.
- Math: $\sqrt{3x} = \sqrt{3} x^{1/2}$
- Rewrite $\sqrt{5x} / x^2$ as $\sqrt{5} x^{(-3/2)}$.

- Math: $\frac{\sqrt{5x}}{x^2} = \sqrt{5}x^{-3/2}$

23. Differentiation rules for exponents

Concept:

- The natural exponential is its own derivative.

Know:

- $d/dx [e^x] = e^x$.
 - Math: $\frac{d}{dx}(e^x) = e^x$
- For a general base, $d/dx [a^x] = a^x \ln a$.
 - Math: $\frac{d}{dx}(a^x) = a^x \ln a$
- For a composite exponential, use the chain rule.
 - Math: $\frac{d}{dx}e^{g(x)} = e^{g(x)}g'(x)$

Practice:

- Differentiate $e^{(2x)}$.
 - Math: e^{2x}
- Answer: $2 e^{(2x)}$.
 - Math: $2e^{2x}$

24. Differentiation rules for logarithms

Concept:

- The derivative of natural log is one over x.

Know:

- $d/dx [\ln x] = 1/x$.
 - Math: $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $d/dx [\ln(g(x))] = g'(x) / g(x)$.
 - Math: $\frac{d}{dx} \ln(g(x)) = \frac{g'(x)}{g(x)}$

Practice:

- Differentiate $\ln(x^2 + 1)$.
 - Math: $\ln(x^2 + 1)$
 - Answer: $2x / (x^2 + 1)$.
 - Math: $\frac{2x}{x^2 + 1}$
-

Part 3: Antiderivatives and integration

25. The anti-derivative, also called the integral

Concept:

- An antiderivative reverses a derivative.
- If $F'(x) = f(x)$, then $F(x)$ is an antiderivative of $f(x)$.
 - Math: if $F'(x) = f(x)$, then $F(x)$ is an antiderivative of $f(x)$.

Know:

- Integration asks: what function gives this derivative?
- There are infinitely many antiderivatives because constants vanish under differentiation.

Practice:

- Find an antiderivative of $3x^2$.
 - Math: $3x^2$
- Answer: $x^3 + C$.
 - Math: $x^3 + C$

26. The power rule for integration

Concept:

- Reverse the power rule: add one to the exponent, then divide by the new exponent.

Know:

- $\int x^n dx = x^{(n+1)} / (n+1) + C$, for n not equal to -1 .
 - Math: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, for $n \neq -1$

Practice:

- Integrate x^4 , $x^{(1/2)}$, and $x^{(-2)}$.
 - Math: x^4 , $x^{1/2}$, x^{-2}
- Answers: $x^5/5 + C$, $(2/3)x^{(3/2)} + C$, $-x^{(-1)} + C$.
 - Math: $\frac{x^5}{5} + C$, $\frac{2}{3}x^{3/2} + C$, $-x^{-1} + C$

27. The power rule for integration will not work for $1/x$

Concept:

- $1/x = x^{(-1)}$, and the power rule would divide by zero.
 - Math: $\frac{1}{x} = x^{-1}$, and $n+1 = 0$ here.

Know:

- $\int 1/x dx = \ln|x| + C$.
 - Math: $\int \frac{1}{x} dx = \ln|x| + C$
- Use the absolute value because the logarithm needs a positive argument.

Practice:

- Integrate $1/x$ and $4/x$.
 - Math: $\frac{1}{x}$ and $\frac{4}{x}$
- Answers: $\ln|x| + C$ and $4 \ln|x| + C$.
 - Math: $\ln|x| + C$ and $4 \ln|x| + C$

28. The constant of integration, plus C

Concept:

- Indefinite integrals always get $+ C$.
- Definite integrals never get $+ C$.

Know:

- $+ C$ represents the whole family of antiderivatives.

- Forgetting + C on an indefinite integral loses the mark.
- Adding + C to a definite integral is also wrong.

Practice:

- Integrate x^2 both ways.
 - Math: x^2
- Indefinite: $x^3/3 + C$.
 - Math: $\int x^2 dx = \frac{x^3}{3} + C$
- Definite from 0 to 1: $1/3$.
 - Math: $\int_0^1 x^2 dx = \frac{1}{3}$

29. Anti-derivative notation

Concept:

- $\int f(x) dx$ means the general antiderivative.
 - Math: $\int f(x) dx$
- $\int [a \text{ to } b] f(x) dx$ means the definite integral.
 - Math: $\int_a^b f(x) dx$

Know:

- The dx tells you the variable of integration.
- The limits a and b appear only on definite integrals.

Practice:

- Write an indefinite integral for $\cos x$.
 - Math: $\int \cos x dx = \sin x + C$
- Write a definite integral for area from 0 to 1.
 - Math: $\int_0^1 f(x) dx$

30. The integral as the area under a curve using the limit

Concept:

- A definite integral is the limit of a sum of rectangle areas.

Know:

- Chop the interval into thin slices.
- Each rectangle has area $f(x) * dx$.
 - Math: $f(x_i^*) \Delta x$
- Add the rectangles and let the width go to 0.
 - Math: $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$

Practice:

- Draw $y = x$ from 0 to 2.
 - Math: $y = x$ on $[0, 2]$
- Approximate the area with four rectangles, then with more.
- See that the limit approaches the triangle area 2.

31. Evaluating definite integrals

Concept:

- Find an antiderivative, then subtract its values at the endpoints.

Know:

- $\int [a \text{ to } b] f(x) \, dx = F(b) - F(a)$.
 - Math: $\int_a^b f(x) \, dx = F(b) - F(a)$
- This is the Fundamental Theorem of Calculus, part 2.

Practice:

- Evaluate $\int [0 \text{ to } 1] x^2 \, dx$.
 - Math: $\int_0^1 x^2 \, dx$
- Answer: $1/3$.
 - Math: $\frac{1}{3}$

32. Definite and indefinite integrals compared

Concept:

- Indefinite integrals return a family of functions.
- Definite integrals return a number.

Know:

- Indefinite: $\int f(x) \, dx = F(x) + C$.
 - Math: $\int f(x) \, dx = F(x) + C$
- Definite: $\int [a \text{ to } b] f(x) \, dx = F(b) - F(a)$.
 - Math: $\int_a^b f(x) \, dx = F(b) - F(a)$

Practice:

- Integrate $2x$ indefinitely, then from 1 to 3.
 - Math: $2x$
- Answers: $x^2 + C$ and 8.
 - Math: $x^2 + C$ and $\int_1^3 2x \, dx = 8$

33. The definite integral and signed area

Concept:

- Area above the x-axis counts positive.
- Area below the x-axis counts negative.
- The definite integral reports signed area.

Know:

- To get true distance or total area, integrate the absolute value, or split at the zeros.
 - Math: total area = $\int_a^b |f(x)| \, dx$

Practice:

- For $v(t) = t^2 - 3t + 2$ on $[0, 3]$, find displacement and distance.
 - Math: $v(t) = t^2 - 3t + 2$ on $[0, 3]$

- See Q19.8.

34. The Fundamental Theorem of Calculus visualized

Concept:

- Differentiation and integration undo each other.

Know:

- FTC part 1: $\frac{d}{dx} \int [a \text{ to } x] f(t) dt = f(x)$.
 - Math: $\frac{d}{dx} \int_a^x f(t) dt = f(x)$
- FTC part 2: $\int [a \text{ to } b] f'(x) dx = f(b) - f(a)$.
 - Math: $\int_a^b f'(x) dx = f(b) - f(a)$
- If the area function grows, its derivative is the height of the curve.

Practice:

- Differentiate $\int [2x \text{ to } x] \sin(t^2) dt$.
 - Math: $\frac{d}{dx} \int_{2x}^x \sin(t^2) dt$
- Answer: $\sin(x^2) - 2 \sin(4x^2)$.
 - Math: $\sin(x^2) - 2 \sin(4x^2)$
- See Q18.1.

35. The integral as a running total of its derivative

Concept:

- An integral accumulates a rate.
- A derivative measures the rate of accumulation.

Know:

- Total water = integral of flow rate.
- Distance = integral of velocity.
- Velocity = derivative of position.
 - Math: $s(b) - s(a) = \int_a^b v(t) dt$ and $v(t) = s'(t)$

Practice:

- Water flows at $r(t) = 180 - 6t$ liters per minute.
 - Math: $r(t) = 180 - 6t$
- Find the amount that flows in the first 15 minutes.
 - Math: $\int_0^{15} (180 - 6t) dt$
- Answer: 2025 L.
 - Math: 2025 litres
- See Q19.3.

36. The trig rule for integration, sine and cosine

Concept:

- Integration flips the derivative signs.

Know:

- $\int \sin x dx = -\cos x + C$.

- Math: $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$.
 - Math: $\int \cos x \, dx = \sin x + C$
- The minus now sits on sine, the opposite of differentiation.

Practice:

- Integrate $\sin x$ and $\cos x$.
 - Math: $\sin x$ and $\cos x$
- Then integrate $x + \cos(2x)$ from a to b .
 - Math: $\int_a^b (x + \cos(2x)) \, dx$

37. Definite integral example problem

Concept:

- A worked example of the full evaluate step.

Know:

- Problem: evaluate $\int [0 \text{ to } \pi] \sin x \, dx$.
 - Math: $\int_0^\pi \sin x \, dx$
- Antiderivative: $-\cos x$.
 - Math: $-\cos x$
- Evaluate: $-\cos(\pi) - (-\cos(0))$.
 - Math: $-\cos(\pi) - (-\cos(0))$
- $-\cos(\pi) = 1$, so the answer is $1 - (-1) = 2$.
 - Math: $-\cos(\pi) = 1$, so the answer is $1 - (-1) = 2$

Practice:

- Repeat this problem without notes.
- Then do $\int [0 \text{ to } \pi/3] \tan x \sec^2 x \, dx$.
 - Math: $\int_0^{\pi/3} \tan x \sec^2 x \, dx$
- Answer: $3/2$.
 - Math: $\frac{3}{2}$

38. u-Substitution

Concept:

- Choose u equal to the inside function.
- Replace dx and the limits, then integrate.

Know:

- You need the derivative of u present, up to a constant.
 - Math: $\int f(g(x))g'(x) \, dx = \int f(u) \, du$ with $u = g(x)$, $du = g'(x) \, dx$
- For definite integrals, change the limits and never convert back.
 - Math: $\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du$

Practice:

- $\int \sin x \cos x \, dx$ with $u = \sin x$.

- Math: $\int \sin x \cos x \, dx$ with $u = \sin x$
- $\int \cos(\sqrt{2x}) / \sqrt{x} \, dx$ with $u = \sqrt{2x}$.
 - Math: $\int \frac{\cos(\sqrt{2x})}{\sqrt{x}} \, dx$ with $u = \sqrt{2x}$
- $\int \sec^3 x \tan x \, dx$ with $u = \sec x$.
 - Math: $\int \sec^3 x \tan x \, dx$ with $u = \sec x$
- See Q16.1 and Q16.2.

39. Integration by parts

Concept:

- Integration by parts reverses the product rule.

Know:

- $\int u \, dv = uv - \int v \, du$.
 - Math: $\int u \, dv = uv - \int v \, du$
- Pick u so that its derivative is simpler.
- Pick dv so that you can integrate it.

Practice:

- Integrate $x e^x \, dx$ with $u = x$ and $dv = e^x \, dx$.
 - Math: $\int x e^x \, dx$ with $u = x, dv = e^x \, dx$
- Integrate $x \cos x \, dx$ with $u = x$ and $dv = \cos x \, dx$.
 - Math: $\int x \cos x \, dx$ with $u = x, dv = \cos x \, dx$

40. The DI method for integration by parts

Concept:

- The DI table organizes repeated integration by parts.

Know:

- Column D holds derivatives of u .
- Column I holds antiderivatives of dv .
- Multiply diagonally, alternating signs, and add the final integral row.

Practice:

- Use a DI table for $x^2 e^x \, dx$.
 - Math: $\int x^2 e^x \, dx$
 - D column: $x^2, 2x, 2, 0$.
 - Math: $x^2, 2x, 2, 0$
 - I column: e^x, e^x, e^x, e^x .
 - Math: e^x, e^x, e^x, e^x
 - Answer: $x^2 e^x - 2x e^x + 2 e^x + C$.
 - Math: $x^2 e^x - 2x e^x + 2 e^x + C$
-

Part 4: Other must-know topics from your exams

These topics are not in your requested list, but they appear repeatedly in the AU sample papers and in your notes.

Functions, domains, and composition

Concept:

- A function gives one output for each input.

Know:

- Vertical line test: a graph is a function if every vertical line hits it at most once.
- Domain excludes zero denominators, negatives under even roots, and non-positive log arguments.
- For $f(g(x))$, keep the domain restrictions of both g and $f(g(x))$.
 - Math: for $f(g(x))$, keep the restrictions of both g and $f \circ g$.

Practice:

- Q01.2, Q01.3, and Q01.4.

Exact trig values

Concept:

- Memorize the unit circle values for 0 , $\pi/6$, $\pi/4$, $\pi/3$, and $\pi/2$.
 - Math: $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$

Know:

- sin: $0, 1/2, \sqrt{2}/2, \sqrt{3}/2, 1$.
 - Math: sin: $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$
- cos: $1, \sqrt{3}/2, \sqrt{2}/2, 1/2, 0$.
 - Math: cos: $1, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, 0$
- tan: $0, 1/\sqrt{3}, 1, \sqrt{3}, \text{undefined}$.
 - Math: tan: $0, \frac{\sqrt{3}}{3}, 1, \sqrt{3}, \text{undefined}$
- Use ASTC for quadrant signs.
- Use sum, difference, and half-angle formulas for 15 and 22.5 degree angles.

The three tools that generate everything outside the five memorized rows:

- Difference: $\cos(A - B) = \cos A \cos B + \sin A \sin B$.
 - Math: $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- Sum: $\sin(A + B) = \sin A \cos B + \cos A \sin B$.
 - Math: $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- Half angle: $\cos(x/2) = \pm \sqrt{(1 + \cos x)/2}$.
 - Math: $\cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}$, sign chosen by the quadrant of $\frac{x}{2}$

Parity, which decides the sign before you look anything up:

- $\cos(-x) = \cos(x)$ and $\sec(-x) = \sec(x)$ are even.
 - Math: $\cos(-x) = \cos x$ and $\sec(-x) = \sec x$
- $\sin(-x) = -\sin(x)$, $\tan(-x) = -\tan(x)$, $\csc(-x) = -\csc(x)$, $\cot(-x) = -\cot(x)$ are odd.
 - Math: $\sin(-x) = -\sin x$, $\tan(-x) = -\tan x$, $\csc(-x) = -\csc x$, $\cot(-x) = -\cot x$

The two tables below are the full reference, carried over from Q01.1. They are split into two five-column tables so each one fits a phone screen held sideways. Note the header marks cos and sec as the even functions, which is the first move on any negative angle.

Do not try to memorize 33 rows. Reduce your angle to a reference angle in $[0^\circ, 90^\circ]$, read the magnitude off the top of the table, then attach the sign from ASTC.

Table 1 of 2 - sine, cosine and tangent

deg	rad	$\sin x$	$\cos x$ (even)	$\tan x$
0	0	0	1	0
15	$\frac{\pi}{12}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$2 - \sqrt{3}$
22.5	$\frac{\pi}{8}$	$\frac{\sqrt{2}-\sqrt{2}}{2}$	$\frac{\sqrt{2}+\sqrt{2}}{2}$	$\sqrt{2} - 1$
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
67.5	$\frac{3\pi}{8}$	$\frac{\sqrt{2}+\sqrt{2}}{2}$	$\frac{\sqrt{2}-\sqrt{2}}{2}$	$\sqrt{2} + 1$
75	$\frac{5\pi}{12}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$2 + \sqrt{3}$
90	$\frac{\pi}{2}$	1	0	undefined
105	$\frac{7\pi}{12}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$-\frac{\sqrt{6}-\sqrt{2}}{4}$	$-(2 + \sqrt{3})$
112.5	$\frac{5\pi}{8}$	$\frac{\sqrt{2}+\sqrt{2}}{2}$	$-\frac{\sqrt{2}-\sqrt{2}}{2}$	$-(\sqrt{2} + 1)$
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-(\sqrt{3})$
135	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-(1)
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
157.5	$\frac{7\pi}{8}$	$\frac{\sqrt{2}-\sqrt{2}}{2}$	$-\frac{\sqrt{2}+\sqrt{2}}{2}$	$-(\sqrt{2} - 1)$
165	$\frac{11\pi}{12}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$-\frac{\sqrt{6}+\sqrt{2}}{4}$	$-(2 - \sqrt{3})$
180	π	0	-(1)	0
195	$\frac{13\pi}{12}$	$-\frac{\sqrt{6}-\sqrt{2}}{4}$	$-\frac{\sqrt{6}+\sqrt{2}}{4}$	$2 - \sqrt{3}$
202.5	$\frac{9\pi}{8}$	$-\frac{\sqrt{2}-\sqrt{2}}{2}$	$-\frac{\sqrt{2}+\sqrt{2}}{2}$	$\sqrt{2} - 1$
210	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
225	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1
240	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
247.5	$\frac{11\pi}{8}$	$-\frac{\sqrt{2}+\sqrt{2}}{2}$	$-\frac{\sqrt{2}-\sqrt{2}}{2}$	$\sqrt{2} + 1$
255	$\frac{17\pi}{12}$	$-\frac{\sqrt{6}+\sqrt{2}}{4}$	$-\frac{\sqrt{6}-\sqrt{2}}{4}$	$2 + \sqrt{3}$
270	$\frac{3\pi}{2}$	-(1)	0	undefined
285	$\frac{19\pi}{12}$	$-\frac{\sqrt{6}+\sqrt{2}}{4}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$-(2 + \sqrt{3})$

deg	rad	$\sin x$	$\cos x$ (even)	$\tan x$
292.5	$\frac{13\pi}{8}$	$-\frac{\sqrt{2+\sqrt{2}}}{2}$	$\frac{\sqrt{2-\sqrt{2}}}{2}$	$-(\sqrt{2} + 1)$
300	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-(\sqrt{3})$
315	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	$-(1)$
330	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
337.5	$\frac{15\pi}{8}$	$-\frac{\sqrt{2-\sqrt{2}}}{2}$	$\frac{\sqrt{2+\sqrt{2}}}{2}$	$-(\sqrt{2} - 1)$
345	$\frac{23\pi}{12}$	$-\frac{\sqrt{6-\sqrt{2}}}{4}$	$\frac{\sqrt{6+\sqrt{2}}}{4}$	$-(2 - \sqrt{3})$
360	2π	0	1	0

Table 2 of 2 - cosecant, secant and cotangent

deg	rad	$\csc x$	$\sec x$ (even)	$\cot x$
0	0	undefined	1	undefined
15	$\frac{\pi}{12}$	$\sqrt{6} + \sqrt{2}$	$\sqrt{6} - \sqrt{2}$	$2 + \sqrt{3}$
22.5	$\frac{\pi}{8}$	$\sqrt{4 + 2\sqrt{2}}$	$\sqrt{4 - 2\sqrt{2}}$	$\sqrt{2} + 1$
30	$\frac{\pi}{6}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45	$\frac{\pi}{4}$	$\sqrt{2}$	$\sqrt{2}$	1
60	$\frac{\pi}{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
67.5	$\frac{3\pi}{8}$	$\sqrt{4 - 2\sqrt{2}}$	$\sqrt{4 + 2\sqrt{2}}$	$\sqrt{2} - 1$
75	$\frac{5\pi}{12}$	$\sqrt{6} - \sqrt{2}$	$\sqrt{6} + \sqrt{2}$	$2 - \sqrt{3}$
90	$\frac{\pi}{2}$	1	undefined	0
105	$\frac{7\pi}{12}$	$\sqrt{6} - \sqrt{2}$	$-(\sqrt{6} + \sqrt{2})$	$-(2 - \sqrt{3})$
112.5	$\frac{5\pi}{8}$	$\sqrt{4 - 2\sqrt{2}}$	$-(\sqrt{4 + 2\sqrt{2}})$	$-(\sqrt{2} - 1)$
120	$\frac{2\pi}{3}$	$\frac{2\sqrt{3}}{3}$	$-(2)$	$-\frac{\sqrt{3}}{3}$
135	$\frac{3\pi}{4}$	$\sqrt{2}$	$-(\sqrt{2})$	$-(1)$
150	$\frac{5\pi}{6}$	2	$-\frac{2\sqrt{3}}{3}$	$-(\sqrt{3})$
157.5	$\frac{7\pi}{8}$	$\sqrt{4 + 2\sqrt{2}}$	$-(\sqrt{4 - 2\sqrt{2}})$	$-(\sqrt{2} + 1)$
165	$\frac{11\pi}{12}$	$\sqrt{6} + \sqrt{2}$	$-(\sqrt{6} - \sqrt{2})$	$-(2 + \sqrt{3})$
180	π	undefined	$-(1)$	undefined
195	$\frac{13\pi}{12}$	$-(\sqrt{6} + \sqrt{2})$	$-(\sqrt{6} - \sqrt{2})$	$2 + \sqrt{3}$
202.5	$\frac{9\pi}{8}$	$-(\sqrt{4 + 2\sqrt{2}})$	$-(\sqrt{4 - 2\sqrt{2}})$	$\sqrt{2} + 1$
210	$\frac{7\pi}{6}$	$-(2)$	$-\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
225	$\frac{5\pi}{4}$	$-(\sqrt{2})$	$-(\sqrt{2})$	1
240	$\frac{4\pi}{3}$	$-\frac{2\sqrt{3}}{3}$	$-(2)$	$\frac{\sqrt{3}}{3}$
247.5	$\frac{11\pi}{8}$	$-(\sqrt{4 - 2\sqrt{2}})$	$-(\sqrt{4 + 2\sqrt{2}})$	$\sqrt{2} - 1$

deg	rad	csc x	sec x (even)	cot x
255	$\frac{17\pi}{12}$	$-(\sqrt{6} - \sqrt{2})$	$-(\sqrt{6} + \sqrt{2})$	$2 - \sqrt{3}$
270	$\frac{3\pi}{2}$	$-(1)$	undefined	0
285	$\frac{19\pi}{12}$	$-(\sqrt{6} - \sqrt{2})$	$\sqrt{6} + \sqrt{2}$	$-(2 - \sqrt{3})$
292.5	$\frac{13\pi}{8}$	$-(\sqrt{4 - 2\sqrt{2}})$	$\sqrt{4 + 2\sqrt{2}}$	$-(\sqrt{2} - 1)$
300	$\frac{5\pi}{3}$	$-\frac{2\sqrt{3}}{3}$	2	$-\frac{\sqrt{3}}{3}$
315	$\frac{7\pi}{4}$	$-(\sqrt{2})$	$\sqrt{2}$	$-(1)$
330	$\frac{11\pi}{6}$	$-(2)$	$\frac{2\sqrt{3}}{3}$	$-(\sqrt{3})$
337.5	$\frac{15\pi}{8}$	$-(\sqrt{4 + 2\sqrt{2}})$	$\sqrt{4 - 2\sqrt{2}}$	$-(\sqrt{2} + 1)$
345	$\frac{23\pi}{12}$	$-(\sqrt{6} + \sqrt{2})$	$\sqrt{6} - \sqrt{2}$	$-(2 + \sqrt{3})$
360	2π	undefined	1	undefined

How to read the two tables:

- The five rows to have cold are 0, 30, 45, 60, 90.
- Everything else is one of those five with a sign change from ASTC, or a member of the 15-degree family or the 22.5-degree family.
- The 15-degree family uses only $\frac{\sqrt{6} + \sqrt{2}}{4}$ and $\frac{\sqrt{6} - \sqrt{2}}{4}$ for sine and cosine, and only $2 + \sqrt{3}$ and $2 - \sqrt{3}$ for tangent.
- The 22.5-degree family uses only $\frac{\sqrt{2 + \sqrt{2}}}{2}$ and $\frac{\sqrt{2 - \sqrt{2}}}{2}$, and only $\sqrt{2} + 1$ and $\sqrt{2} - 1$.
- Table 2 is Table 1 turned upside down, sign and all, because a reciprocal keeps the sign of what it came from.
- $|\csc|$ and $|\sec|$ are never less than 1, which is a free sanity check.
- $\tan$ and $\sec$ are undefined at 90° and 270° , where cosine is zero.
- $\cot$ and $\csc$ are undefined at 0° , 180° and 360° , where sine is zero.
- Tangent and cotangent repeat every 180° with no sign change, which is why 15° and 195° share $\tan = 2 - \sqrt{3}$.
- For an angle outside $[0^\circ, 360^\circ]$, add or subtract full turns first: $\frac{25\pi}{12} = 375^\circ$ is coterminal with 15° .

Practice:

- Q01.1.

Graph transformations

Concept:

- Transform a parent graph in a fixed order.

Know:

- For $y = a f(b(x - h)) + k$, work inside out.
 - Math: $y = a f(b(x - h)) + k$
- Horizontal shift first, then horizontal stretch or reflection.
- Then vertical stretch, then vertical reflection, then vertical shift.
- The horizontal shift feels backwards: $(x + 4)$ means left 4.
 - Math: $(x + 4)$ means a shift left by 4

Practice:

- Q02.1 and Q02.2.

Limits and continuity

Concept:

- A function is continuous at a point if the limit equals the function value.

Know:

- Continuity needs three things: $f(a)$ exists, the limit exists, and they are equal.
 - Math: $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and $\lim_{x \rightarrow a} f(x) = f(a)$
- Differentiable implies continuous, but not the reverse.
- $|x|$ is continuous but not differentiable at 0.
 - Math: $|x|$ is continuous at 0 but not differentiable there.

Practice:

- Q03.1, Q03.2, Q03.3.

Asymptotes

Concept:

- Vertical asymptotes come from remaining denominator zeros.
- Horizontal asymptotes come from limits at infinity.

Know:

- Factor and cancel first.
- A canceled factor is a hole, not an asymptote.
- For a rational function, compare degrees for the horizontal asymptote.
 - Math: if deg numerator < deg denominator then $y = 0$; if equal then y is the ratio of leading coefficients; if greater then there is no horizontal asymptote.

Practice:

- Q04.1.

Tangent lines and perpendicular slopes

Concept:

- The slope of the tangent line at a is $f'(a)$.
 - Math: the slope at $x = a$ is $f'(a)$

Know:

- Parallel means equal slopes.
- Perpendicular means negative reciprocal slopes.
- Perpendicular to slope m means $f'(x) = -1/m$.
 - Math: $f'(x) = -\frac{1}{m}$
- Point-slope form of the tangent line: $y - f(a) = f'(a)(x - a)$.
 - Math: $y - f(a) = f'(a)(x - a)$

Practice:

- Q09.1.

Implicit differentiation

Concept:

- Differentiate both sides with respect to x .
- Every y produces a y' by the chain rule.

$$\text{- Math: } \frac{d}{dx}(y^n) = n y^{n-1} y'$$

Know:

- Products of x and y need the product rule.

$$\text{- Math: } \frac{d}{dx}(xy) = y + xy'$$

- Collect all y' terms, factor, and solve.

Practice:

- Q10.1 and Q10.2.

Related rates

Concept:

- Variables change with time.
- Differentiate an equation with respect to t .
 - Math: differentiate with respect to t

Know:

- Name the variables and write what is given and wanted as derivatives with respect to t .
 - Math: given $\frac{dA}{dt}$, wanted $\frac{dB}{dt}$
- Eliminate variables you have no rate for before differentiating.
- Substitute numbers only after differentiating.

Practice:

- Gravel cone: Q11.1.
- Rocket and radar: Q11.2.

Linear approximation and differentials

Concept:

- Use the tangent line to estimate nearby values.

Know:

- $f(a + dx) \approx f(a) + f'(a) dx$.
 - Math: $f(a + dx) \approx f(a) + f'(a) dx$
- Choose a as the nearest value you know exactly.
- Convert degrees to radians for trig problems.
 - Math: $1^\circ = \frac{\pi}{180}$ radians

Practice:

- Q12.1, Q12.2, and Q12.3.

Newton's method

Concept:

- Approximate a root with repeated tangent-line steps.

Know:

- $x_{n+1} = x_n - f(x_n) / f'(x_n)$.
 - Math: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
- Keep all decimals between steps and round only at the end.

Practice:

- Q13.1.

Curve sketching with calculus

Concept:

- Use domain, asymptotes, f' , and f'' to build an accurate graph.
 - Math: use f' and f''

Know:

- $f' = 0$ gives critical points.
 - Math: $f'(x) = 0$
- f' undefined while f is defined gives a cusp or vertical tangent.
- Sign of f' gives increasing and decreasing.
- Sign of f'' gives concavity and inflection points.

Practice:

- Q14.1.

Extreme values on a closed interval

Concept:

- On a closed interval, absolute extrema happen at critical numbers or endpoints.

Know:

- List all critical numbers where $f' = 0$ or f' is undefined.
 - Math: $f'(c) = 0$ or $f'(c)$ does not exist
- Evaluate f at those numbers and at both endpoints.
- The largest value is the absolute max and the smallest is the absolute min.

Practice:

- Q15.1, Q15.3, and Q15.5.

Optimization

Concept:

- Convert a word problem into a one-variable function and optimize it.

Know:

- Fixed area and least perimeter give a square.
- Fixed volume and cheapest cost require including all differently priced surfaces.

Practice:

- Q15.2 and Q15.4.

Properties of the definite integral and bounding

Concept:

- Bounds on a function produce bounds on its integral.

Know:

- If $m \leq f(x) \leq M$ on $[a, b]$, then $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$.
 - Math: if $m \leq f(x) \leq M$ on $[a, b]$, then $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$
- The width is $b - a$, not just b .
 - Math: the width is $b - a$

Practice:

- Q17.1, Q17.2, and Q17.3.

Area between curves

Concept:

- Integrate upper curve minus lower curve.

Know:

- Find intersection points by setting the curves equal.
 - Math: $A = \int_a^b (y_{\text{top}} - y_{\text{bottom}}) dx$
- If the curves cross inside the interval, split the integral.

Practice:

- Q19.1 and Q19.2.

Net change and motion

Concept:

- Net change equals the integral of a rate.

Know:

- Displacement is $\int v \, dt$.
 - Math: displacement $= \int_a^b v(t) \, dt$
- Distance is $\int |v| \, dt$.
 - Math: distance $= \int_a^b |v(t)| \, dt$
- Split distance at the zeros of velocity.

Practice:

- Q19.3, Q19.8, and Q19.9.

Work

Concept:

- Work equals force integrated over distance.

Know:

- Spring: $F = kx$, so $W = (1/2) k d^2$.

- Math: $F = kx$, so $W = \int_0^d kx \, dx = \frac{1}{2}kd^2$
- Chain: integrate the weight of each small piece over the height it is lifted.
 - Math: $W = \int \rho g y \, dy$

Practice:

- Q19.4 and Q19.5.

Average value of a function

Concept:

- Average value is the integral divided by the interval width.

Know:

- $f_{\text{avg}} = (1 / (b - a)) \int [a \text{ to } b] f(x) \, dx$.
- Math: $f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx$

Practice:

- Q19.6 and Q19.7.
-

Part 5: Every exam question, restated and solved

This part restates all 49 questions from the question bank notes\MATH265.md in ordinary mathematical symbols, then works each one step by step.

How each entry is laid out:

- **Question** is the question restated in mathematical symbols.
- **Steps** are the working, with the reason for each move.
- **Answer** is the result in mathematical symbols.
- **Möbius** is the same result flattened into the text you type into the answer box.

Every numeric and symbolic result below was independently recomputed in Python with sympy before it was written down. Where a sketch is the answer, the entry describes exactly what the sketch must show, because a machine cannot mark a drawing and a human marker will look for those features.

1. Functions and precalculus foundations

Q1.1 - Exact value of a cosine

Question. Give the exact value of $\cos\left(-\frac{\pi}{12}\right)$.

Step 1: kill the minus sign. Cosine is even, so $\cos(-x) = \cos x$.

$$\cos\left(-\frac{\pi}{12}\right) = \cos\left(\frac{\pi}{12}\right)$$

Step 2: recognise the angle. $\frac{\pi}{12} = 15^\circ$, which is not one of the five memorized angles, but it is $45^\circ - 30^\circ$.

Step 3: apply the cosine difference formula. The formula is $\cos(A - B) = \cos A \cos B + \sin A \sin B$. Note the sign flips: a minus inside gives a plus outside.

$$\cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6}$$

Step 4: substitute the five memorized values.

$$= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}$$

Step 5: combine over one denominator.

Answer.

$$\cos\left(-\frac{\pi}{12}\right) = \frac{\sqrt{6} + \sqrt{2}}{4} \approx 0.9659258263$$

Möbius: (sqrt(6)+sqrt(2))/4

Trap. Typing sqrt(6)+sqrt(2)/4 parses as $\sqrt{6} + \frac{\sqrt{2}}{4} \approx 2.80$, which is not even in range for a cosine. Bracket the whole numerator.

Q1.2 - Composite functions and their domains

Question. Let $f(x) = 2x^2 - 5$ and $g(x) = \frac{x+5}{2x-9}$. Find the composite functions and their domains.

- a. $f(g(x))$
- b. $g(f(x))$

Part a, step 1: substitute g into f . Everywhere f has an x , write the whole of $g(x)$.

$$f(g(x)) = 2\left(\frac{x+5}{2x-9}\right)^2 - 5$$

Part a, step 2: expand if you want a single fraction.

$$= \frac{2(x+5)^2}{(2x-9)^2} - 5 = \frac{2(x+5)^2 - 5(2x-9)^2}{(2x-9)^2}$$

Part a, step 3: the domain. The domain of a composite keeps the restrictions of the inner function *and* of the result. Here g needs $2x - 9 \neq 0$, and f itself restricts nothing because a polynomial accepts every real number.

$$2x - 9 \neq 0 \implies x \neq \frac{9}{2}$$

Answer a.

$$f(g(x)) = 2\left(\frac{x+5}{2x-9}\right)^2 - 5, \quad \text{domain } \left(-\infty, \frac{9}{2}\right) \cup \left(\frac{9}{2}, \infty\right)$$

Möbius: $2*((x+5)/(2x-9))^2 - 5$, domain $x \neq 9/2$

Part b, step 1: substitute f into g .

$$g(f(x)) = \frac{(2x^2 - 5) + 5}{2(2x^2 - 5) - 9}$$

Part b, step 2: simplify. The numerator collapses because $-5 + 5 = 0$, and the denominator is $4x^2 - 10 - 9$.

$$= \frac{2x^2}{4x^2 - 19}$$

Part b, step 3: the domain. The inner function f is a polynomial, so it restricts nothing. The result needs a non-zero denominator.

$$4x^2 - 19 \neq 0 \implies x^2 \neq \frac{19}{4} \implies x \neq \pm \frac{\sqrt{19}}{2}$$

Answer b.

$$g(f(x)) = \frac{2x^2}{4x^2 - 19}, \quad \text{domain } x \neq \pm \frac{\sqrt{19}}{2}$$

Möbius: $2x^2/(4x^2-19)$, domain $x \neq \sqrt{19}/2$ and $x \neq -\sqrt{19}/2$

Trap. The numerator simplifying to $2x^2$ tempts you to say the domain is all reals. It is not: the denominator still bans $\pm \frac{\sqrt{19}}{2}$.

Q1.3 - Which equations define a function of x

Question. Determine which of the following equations are well defined functions with independent variable x , and explain.

- a. $\frac{2y}{x^2 - 3|x|} = 1$
- b. $y^2 + x^2 = 10$

Part a, step 1: solve for y . Multiply both sides by the denominator.

$$2y = x^2 - 3|x| \implies y = \frac{x^2 - 3|x|}{2}$$

Part a, step 2: test the definition. Each x now produces exactly one y , because the right-hand side is a single arithmetic expression with no $\pm$. So this **is** a function.

Part a, step 3: state the domain. The original equation had a denominator, and that restriction survives.

$$x^2 - 3|x| = 0 \implies |x|(|x| - 3) = 0 \implies x = 0, x = 3, x = -3$$

Answer a. Yes, it is a function of x :

$$y = \frac{x^2 - 3|x|}{2}, \quad \text{domain } x \neq 0, x \neq \pm 3$$

Möbius: $(x^2 - 3 \cdot \text{abs}(x))/2$, domain $x \neq 0, x \neq 3, x \neq -3$

Part b, step 1: solve for y .

$$y^2 = 10 - x^2 \implies y = \pm \sqrt{10 - x^2}$$

Part b, step 2: test the definition. The $\pm$ is fatal. At $x = 1$ the equation gives both $y = 3$ and $y = -3$, so one input has two outputs.

Part b, step 3: the graphical reason. $x^2 + y^2 = 10$ is a circle of radius $\sqrt{10}$ centred at the origin, and a vertical line through $x = 1$ cuts it twice, so it fails the vertical line test.

Answer b. No, it is **not** a function of x , because $y = \pm \sqrt{10 - x^2}$ assigns two values of y to each x in $(-\sqrt{10}, \sqrt{10})$.

Möbius: not a function; the relation is $x^2 + y^2 = 10$, a circle of radius $\sqrt{10}$

Q1.4 - Volume of a cone as a function of its radius

Question. A cone has height double its radius, so $h = 2r$. The volume of a cone is $V = \frac{\pi r^2 h}{3}$.

- a. Express V as a function of r .
- b. Find the domain.
- c. Compute V for $r = 3$.

Part a, step 1: write the constraint. "Height is double the radius" is $h = 2r$.

Part a, step 2: eliminate h . Substitute $h = 2r$ into the volume formula, so only r remains.

$$V(r) = \frac{\pi r^2 (2r)}{3} = \frac{2\pi r^3}{3}$$

Answer a.

$$V(r) = \frac{2\pi r^3}{3}$$

Möbius: $2\pi r^3/3$

Part b: the domain. Algebraically $\frac{2\pi r^3}{3}$ accepts every real number, but a radius is a physical length. A cone with radius 0 or negative radius is not a cone, so the model requires a strictly positive radius.

Answer b.

$$r > 0, \quad \text{that is } (0, \infty)$$

Möbius: $r > 0$

Part c: evaluate at $r = 3$.

$$V(3) = \frac{2\pi(3)^3}{3} = \frac{2\pi \cdot 27}{3} = 18\pi$$

Answer c.

$$V(3) = 18\pi \approx 56.55 \text{ cubic units}$$

Möbius: 18π

Trap. Leaving the answer as $\frac{54\pi}{3}$ is unsimplified, and answering 56.55 when the question did not ask for a decimal throws away the exact form.

2. Graph transformations

Q2.1 - Graph a shifted parabola by transformations

Question. Give a labeled graph of $g(x) = 3(x + 4)^2$ by starting from a basic function and applying transformations, explaining the procedure. No credit for any other method.

Step 1: name the parent function.

$$y = x^2$$

This is the basic parabola, vertex at $(0, 0)$, opening upward, passing through $(\pm 1, 1)$ and $(\pm 2, 4)$.

Step 2: match the general form. Compare against $y = a f(b(x - h)) + k$.

$$g(x) = 3(x + 4)^2 \text{ gives } a = 3, b = 1, h = -4, k = 0$$

Step 3: horizontal shift, first. $(x + 4)$ means $h = -4$, so the graph moves **left 4**, not right. The shift is inside the function, and inside operations run backwards.

$$y = x^2 \longrightarrow y = (x + 4)^2, \quad \text{vertex } (0, 0) \rightarrow (-4, 0)$$

Step 4: vertical stretch, second. The factor $a = 3$ is outside, so it multiplies every output by 3. This makes the parabola three times steeper, and it does **not** move the vertex, because $3 \times 0 = 0$.

$$y = (x + 4)^2 \longrightarrow y = 3(x + 4)^2$$

Step 5: no reflection and no vertical shift. $a = 3$ is positive, so nothing flips. $k = 0$, so nothing moves up or down.

Answer. Order of transformations: **shift left 4, then stretch vertically by 3.**

Key points to label on the sketch:

$$\text{vertex } (-4, 0), \quad (-3, 3), \quad (-5, 3), \quad (-2, 12), \quad (-6, 12)$$

The curve opens upward, is symmetric about the vertical line $x = -4$, and its axis of symmetry should be drawn and labeled.

Möbius: vertex $(-4, 0)$, axis of symmetry $x = -4$, function $3(x + 4)^2$

Trap. Shifting right 4 because the sign says $+4$. Set the bracket to zero: $x + 4 = 0$ gives $x = -4$, which is where the vertex lands.

Q2.2 - Graph a reflected absolute value by transformations

Question. State the transformations, in order, applied to the basic graph of $|x|$ to obtain $f(x) = -3|x-2|$, then sketch. No credit for any other method.

Step 1: name the parent function.

$$y = |x|$$

A V shape with its corner at $(0,0)$, slope -1 on the left and $+1$ on the right.

Step 2: match the general form.

$$f(x) = -3|x-2| \quad \text{gives} \quad a = -3, \quad b = 1, \quad h = 2, \quad k = 0$$

Step 3: horizontal shift, first. $(x-2)$ means $h = +2$, so the graph moves **right 2**.

$$y = |x| \longrightarrow y = |x-2|, \quad \text{corner } (0,0) \rightarrow (2,0)$$

Step 4: vertical stretch by 3, second. The magnitude of a is 3, so outputs triple and the V becomes narrower.

$$y = |x-2| \longrightarrow y = 3|x-2|$$

Step 5: reflection in the x -axis, third. The sign of a is negative, so the whole graph flips over the x -axis and the V now opens **downward**.

$$y = 3|x-2| \longrightarrow y = -3|x-2|$$

Step 6: no vertical shift. $k = 0$, so the corner stays on the x -axis.

Answer. Order: **shift right 2, stretch vertically by 3, reflect in the x -axis.**

Key points to label:

$$\text{corner } (2,0), \quad (1,-3), \quad (3,-3), \quad (0,-6), \quad (4,-6)$$

The corner at $(2,0)$ is the **maximum**, the arms have slopes $+3$ on the left and -3 on the right, and the function is continuous everywhere but not differentiable at $x = 2$.

Möbius: corner $(2,0)$, function $-3 \cdot \text{abs}(x-2)$

Trap. Reflecting before stretching gives the same picture here, but reflecting before *shifting* does not. Always do the inside horizontal move first.

3. Limits and continuity**Q3.1 - Five limits, algebraic and trigonometric**

Question. Evaluate each limit; if it does not exist, explain why.

- a. $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$
- b. $\lim_{x \rightarrow -2} \frac{3x^2 - 2x - 16}{(x+2)^2}$
- c. $\lim_{x \rightarrow 3} \frac{2x^2 - x + 1}{x-3}$

- d. $\lim_{x \rightarrow 2} \frac{\sqrt{6-x}-2}{\sqrt{3-x}-1}$
- e. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x^2-x}$

Part a. A polynomial is continuous everywhere, so direct substitution is legal.

$$3(2)^2 - 2(2) + 1 = 12 - 4 + 1 = 9$$

Answer a. 9. **Möbius:** 9

Part b, step 1: substitute and diagnose. Numerator at $x = -2$: $3(4) - 2(-2) - 16 = 12 + 4 - 16 = 0$. Denominator: 0. So this is $\frac{0}{0}$ and must be rewritten.

Part b, step 2: factor the numerator. The roots of $3x^2 - 2x - 16$ are $x = \frac{2 \pm 14}{6}$, that is $\frac{8}{3}$ and -2 .

$$3x^2 - 2x - 16 = (x + 2)(3x - 8)$$

Part b, step 3: cancel one factor.

$$\frac{(x+2)(3x-8)}{(x+2)^2} = \frac{3x-8}{x+2}$$

Part b, step 4: re-diagnose. Now the numerator tends to $3(-2) - 8 = -14$, which is not zero, while the denominator tends to 0. This is the $\frac{k}{0}$ case, so check each side.

$$x \rightarrow -2^- : x + 2 \rightarrow 0^-, \frac{-14}{0^-} \rightarrow +\infty$$

$$x \rightarrow -2^+ : x + 2 \rightarrow 0^+, \frac{-14}{0^+} \rightarrow -\infty$$

Answer b. The limit **does not exist**, because the left limit is $+\infty$ and the right limit is $-\infty$, so the two one-sided limits disagree.

Möbius: DNE

Part c, step 1: substitute and diagnose. Numerator: $2(9) - 3 + 1 = 16$, not zero. Denominator: 0. This is $\frac{k}{0}$ with $k \neq 0$, so there is nothing to cancel; go straight to the sides.

$$x \rightarrow 3^- : \frac{16}{0^-} \rightarrow -\infty \quad x \rightarrow 3^+ : \frac{16}{0^+} \rightarrow +\infty$$

Answer c. The limit **does not exist**; the function drops to $-\infty$ on the left of 3 and rises to $+\infty$ on the right, so $x = 3$ is a vertical asymptote with opposite behaviour on the two sides.

Möbius: DNE

Part d, step 1: substitute and diagnose. Numerator: $\sqrt{4} - 2 = 0$. Denominator: $\sqrt{1} - 1 = 0$. This is $\frac{0}{0}$, and the presence of roots says rationalize.

Part d, step 2: multiply by both conjugates at once. Multiply top and bottom by $(\sqrt{6-x} + 2)$ and by $(\sqrt{3-x} + 1)$.

The numerator conjugate product is $(6-x)-4 = 2-x$. The denominator conjugate product is $(3-x)-1 = 2-x$.

$$\frac{\sqrt{6-x}-2}{\sqrt{3-x}-1} = \frac{(2-x)(\sqrt{3-x}+1)}{(2-x)(\sqrt{6-x}+2)}$$

Part d, step 3: cancel the common $(2-x)$. This is the whole trick: the same factor $2-x$ appears top and bottom.

$$= \frac{\sqrt{3-x}+1}{\sqrt{6-x}+2}$$

Part d, step 4: now substitute.

$$= \frac{\sqrt{1}+1}{\sqrt{4}+2} = \frac{2}{4} = \frac{1}{2}$$

Answer d. $\frac{1}{2}$. **Möbius:** 1/2

Part e, step 1: diagnose. $\sin(0) = 0$ and $0^2 - 0 = 0$, so it is $\frac{0}{0}$, and the sin says use the special limit.

Part e, step 2: factor the denominator and build the special limit. The special limit is $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$, so force a $3x$ under the $\sin(3x)$.

$$\frac{\sin(3x)}{x^2 - x} = \frac{\sin(3x)}{x(x-1)} = \frac{\sin(3x)}{3x} \cdot \frac{3x}{x(x-1)} = \frac{\sin(3x)}{3x} \cdot \frac{3}{x-1}$$

Part e, step 3: take the limit of each factor.

$$\rightarrow 1 \cdot \frac{3}{0-1} = -3$$

Answer e. -3 . **Möbius:** -3

Q3.2 - Six limits, with justification required

Question. Evaluate each limit; if it does not exist, explain why. No credit for unjustified answers.

- a. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1}$
- b. $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos(2x)}$
- c. $\lim_{x \rightarrow 0} \frac{6x - 9}{x^3 - 12x + 3}$
- d. $\lim_{x \rightarrow \infty} \frac{5 - 2x^3}{x^2 + 2}$
- e. $\lim_{x \rightarrow -\pi/3} \frac{\tan(2x)}{3x + \pi}$

• f. $\lim_{x \rightarrow 2} \frac{\cos(\pi x)}{(x-2)^2}$

Part a, step 1: diagnose. Both $x^2 - 1$ and $x^3 - 1$ vanish at $x = 1$, so it is $\frac{0}{0}$ and $(x - 1)$ is a common factor.

Part a, step 2: factor both.

$$x^2 - 1 = (x - 1)(x + 1), \quad x^3 - 1 = (x - 1)(x^2 + x + 1)$$

Part a, step 3: cancel and substitute.

$$\frac{(x-1)(x+1)}{(x-1)(x^2+x+1)} = \frac{x+1}{x^2+x+1} \rightarrow \frac{2}{3}$$

Answer a. $\frac{2}{3}$. **Möbius:** 2/3

Part b, step 1: diagnose. $1 - \cos 0 = 0$ and $0^2 = 0$, so $\frac{0}{0}$.

Part b, step 2: replace the cosine with an identity. Use $1 - \cos(2\theta) = 2 \sin^2 \theta$ with $\theta = x$. This is cleaner than the half-angle route because it removes the cosine entirely.

$$\frac{x^2}{1 - \cos(2x)} = \frac{x^2}{2 \sin^2 x}$$

Part b, step 3: build the special limit.

$$= \frac{1}{2} \left(\frac{x}{\sin x} \right)^2 \rightarrow \frac{1}{2} (1)^2 = \frac{1}{2}$$

Answer b. $\frac{1}{2}$. **Möbius:** 1/2

Part c, step 1: substitute. This one is not indeterminate at all, which is the point of including it.

$$\frac{6(0) - 9}{0 - 0 + 3} = \frac{-9}{3} = -3$$

Answer c. -3. **Möbius:** -3

Part d, step 1: diagnose. Both parts grow without bound, so it is $\frac{\infty}{\infty}$.

Part d, step 2: divide by the highest power in the denominator, x^2 .

$$\frac{5 - 2x^3}{x^2 + 2} = \frac{\frac{5}{x^2} - 2x}{1 + \frac{2}{x^2}}$$

Part d, step 3: take the limit. The numerator behaves like $-2x \rightarrow -\infty$ and the denominator tends to 1.

Answer d.

$$\lim_{x \rightarrow \infty} \frac{5 - 2x^3}{x^2 + 2} = -\infty$$

The degree of the numerator exceeds the degree of the denominator, so there is no horizontal asymptote, and the negative leading coefficient sends it to $-\infty$.

Möbius: -infinity

Part e, step 1: substitute carefully. The denominator $3x + \pi \rightarrow 3\left(-\frac{\pi}{3}\right) + \pi = 0$. For the numerator, $2x \rightarrow -\frac{2\pi}{3}$, and

$$\tan\left(-\frac{2\pi}{3}\right) = -\tan\left(\frac{2\pi}{3}\right) = -(-\sqrt{3}) = \sqrt{3}$$

Part e, step 2: diagnose. This is $\frac{\sqrt{3}}{0}$, not $\frac{0}{0}$, so there is nothing to cancel. Check the sides.

$$x \rightarrow \left(-\frac{\pi}{3}\right)^- : 3x + \pi \rightarrow 0^-, \quad \frac{\sqrt{3}}{0^-} \rightarrow -\infty$$

$$x \rightarrow \left(-\frac{\pi}{3}\right)^+ : 3x + \pi \rightarrow 0^+, \quad \frac{\sqrt{3}}{0^+} \rightarrow +\infty$$

Answer e. The limit **does not exist**, because the numerator tends to $\sqrt{3} \neq 0$ while the denominator tends to 0 through both signs, giving $-\infty$ from the left and $+\infty$ from the right.

Möbius: DNE

Trap. This looks like it should be a $\frac{0}{0}$ tangent limit, so people reach for $\frac{\tan u}{u} \rightarrow 1$. Check the numerator first: $\tan\left(-\frac{2\pi}{3}\right) = \sqrt{3}$, not 0, so that tool does not apply.

Part f, step 1: substitute. Numerator: $\cos(2\pi) = 1$. Denominator: $(2 - 2)^2 = 0$, and because it is squared it approaches 0 from the **positive** side only.

Part f, step 2: read the sign.

$$\frac{1}{0^+} \rightarrow +\infty \quad \text{from both sides}$$

Answer f.

$$\lim_{x \rightarrow 2} \frac{\cos(\pi x)}{(x - 2)^2} = +\infty$$

The limit does not exist as a finite number, but because the squared denominator is positive on both sides the function goes to $+\infty$ from both directions, so we may write the limit as $+\infty$.

Möbius: infinity

Q3.3 - Two limits, at infinity and at an asymptote**Question.** Evaluate; no credit for unjustified answers.

- a. $\lim_{x \rightarrow \infty} \left(\frac{3x^2 - 4}{4 + 2x + 2x^4} \right) \sin x$
- b. $\lim_{x \rightarrow 5\pi/2} \tan x$

Part a, step 1: split the product. The $\sin x$ has no limit at infinity, so do not try to evaluate the product directly. Handle the rational factor on its own.**Part a, step 2: limit of the rational factor.** The denominator has degree 4 and the numerator degree 2, so the fraction dies.

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 4}{2x^4 + 2x + 4} = \lim_{x \rightarrow \infty} \frac{\frac{3}{x^2} - \frac{4}{x^4}}{2 + \frac{2}{x^3} + \frac{4}{x^4}} = \frac{0}{2} = 0$$

Part a, step 3: bound the oscillating factor.

$$-1 \leq \sin x \leq 1 \quad \text{for all } x$$

Part a, step 4: apply the Squeeze Theorem. Let $R(x)$ be the rational factor. Then

$$-|R(x)| \leq R(x) \sin x \leq |R(x)|$$

Both outer bounds tend to 0, so the middle is squeezed to 0.

Answer a. 0, by the Squeeze Theorem: a bounded factor times a factor going to zero goes to zero.**Möbius:** 0**Part b, step 1: locate the angle.**

$$\frac{5\pi}{2} = 2\pi + \frac{\pi}{2}$$

so this is coterminal with $\frac{\pi}{2}$, where $\cos x = 0$ and $\tan x$ is undefined.**Part b, step 2: check both sides.** Write $\tan x = \frac{\sin x}{\cos x}$, with $\sin\left(\frac{5\pi}{2}\right) = 1$.

$$x \rightarrow \left(\frac{5\pi}{2}\right)^- : \cos x \rightarrow 0^+, \quad \tan x \rightarrow +\infty$$

$$x \rightarrow \left(\frac{5\pi}{2}\right)^+ : \cos x \rightarrow 0^-, \quad \tan x \rightarrow -\infty$$

Answer b. The limit **does not exist**, because $\tan x \rightarrow +\infty$ from the left and $\tan x \rightarrow -\infty$ from the right; $x = \frac{5\pi}{2}$ is a vertical asymptote of the tangent.**Möbius:** DNE

4. Asymptotes

Q4.1 - Vertical and horizontal asymptotes of a rational function

Question. Find the vertical and horizontal asymptotes of

$$f(x) = \frac{x^2 - 3x - 4}{x^2 - 16}$$

Step 1: factor both parts before doing anything else. This is the whole question; unfactored, you will report an asymptote that is not there.

$$x^2 - 3x - 4 = (x - 4)(x + 1), \quad x^2 - 16 = (x - 4)(x + 4)$$

Step 2: note the original domain. Before cancelling, the denominator bans $x = 4$ and $x = -4$.

$$f(x) = \frac{(x - 4)(x + 1)}{(x - 4)(x + 4)}$$

Step 3: cancel, and record what the cancellation means.

$$f(x) = \frac{x + 1}{x + 4}, \quad x \neq 4$$

A cancelled factor is a **hole**, not an asymptote. The hole sits at

$$x = 4, \quad y = \frac{4 + 1}{4 + 4} = \frac{5}{8}$$

Step 4: vertical asymptote from the surviving denominator zero.

$$x + 4 = 0 \implies x = -4$$

Confirm it really blows up, since the numerator there is $-4 + 1 = -3 \neq 0$.

$$x \rightarrow -4^- : \frac{-3}{0^-} \rightarrow +\infty \quad x \rightarrow -4^+ : \frac{-3}{0^+} \rightarrow -\infty$$

Step 5: horizontal asymptote by comparing degrees. Numerator and denominator of the original both have degree 2, so the limit is the ratio of leading coefficients.

$$\lim_{x \rightarrow \pm\infty} \frac{x^2 - 3x - 4}{x^2 - 16} = \frac{1}{1} = 1$$

Answer.

Vertical asymptote: $x = -4$ Horizontal asymptote: $y = 1$

Hole at $\left(4, \frac{5}{8}\right)$, which is not an asymptote

Möbius: VA $x = -4$, HA $y = 1$, hole at $(4, 5/8)$

Trap. Reporting $x = 4$ as a vertical asymptote. The factor $(x - 4)$ cancels, so the graph has a removable discontinuity there, a single missing point, not a blow-up.

5. Graphical interpretation and sketching from conditions

These three questions are marked on the *features* of your sketch, so the answer below lists exactly what a marker looks for.

Q5.1 - Sketch a function from four conditions

Question. Sketch the graph of a single function $f(x)$ satisfying all of:

- $\lim_{x \rightarrow 0} f(x)$ does not exist
- $f(0) = 0$
- $\lim_{x \rightarrow \infty} f(x) = -1$
- f is not differentiable at $x = -2$

Condition 1: the limit at 0 does not exist. Meaning: the left and right pieces approach different heights, so the graph **jumps** at $x = 0$. Draw a jump discontinuity: for instance the left piece rising to an open circle at $(0, 2)$ and the right piece starting at an open circle at $(0, -3)$.

Condition 2: $f(0) = 0$. Meaning: despite the jump, the function is defined at 0, and its value is exactly 0. Draw a **filled dot at the origin** $(0, 0)$, sitting on neither branch. This is the condition students drop; the two open circles need a third, solid point at $(0, 0)$.

Condition 3: $\lim_{x \rightarrow \infty} f(x) = -1$. Meaning: far to the right the graph flattens onto the horizontal line $y = -1$. Draw the dashed asymptote $y = -1$ and have the right branch approach it without crossing repeatedly.

Condition 4: not differentiable at $x = -2$. Meaning: the graph exists there but has no single tangent. The cheapest legal choice is a **sharp corner** at $x = -2$, like the vertex of a V. A cusp or a vertical tangent would also satisfy it; a hole or jump would too, but a corner is easiest to draw convincingly.

Answer: the checklist your sketch must show.

jump at $x = 0$; solid dot $(0, 0)$; dashed $y = -1$ as $x \rightarrow \infty$; corner at $x = -2$

Möbius: sketch question, no typed answer; the four features above are the marks.

Trap. Making $x = -2$ a hole *and* calling it non-differentiable is technically fine but weaker, since a marker wants to see you know a corner is continuous-but-not-differentiable.

Q5.2 - Interpret five conditions, then sketch

Question. Interpret each condition in terms of the graph of f , then sketch one function satisfying all of them.

- a. $\lim_{x \rightarrow \infty} f(x) = 3$
- b. $f(0) = 0$
- c. $\lim_{x \rightarrow -1} f(x) = 2$
- d. $\lim_{x \rightarrow 1} f(x) = \infty$
- e. f is continuous but not differentiable at $x = 0$

Interpretation a. As x grows without bound the outputs settle at height 3. Graphically: a **horizontal asymptote** $y = 3$ on the right.

Interpretation b. The graph passes through the origin. Graphically: the point $(0, 0)$ is on the curve.

Interpretation c. Approaching $x = -1$ from either side the outputs approach 2. Note the condition says nothing about $f(-1)$ itself, so the point may be filled at $(-1, 2)$, or a hole with the actual value elsewhere. Graphically: the curve funnels to height 2 at $x = -1$.

Interpretation d. As x approaches 1 the outputs increase without bound on both sides. Graphically: a **vertical asymptote** $x = 1$, with the curve going to $+\infty$ on *both* sides.

Interpretation e. Continuous at 0 means no break, so the limit exists and equals $f(0) = 0$. Not differentiable means no unique tangent line there. Graphically: a **corner** at the origin, like $|x|$.

Answer: the checklist your sketch must show.

dashed $y = 3$ on the right; through $(0,0)$ with a corner; value 2 at $x = -1$; dashed $x = 1$ with $+\infty$ both sides

A workable shape: a V-corner at the origin, the left arm passing through height 2 at $x = -1$, the branch just left of $x = 1$ climbing to $+\infty$, and a separate branch for $x > 1$ dropping from $+\infty$ and flattening onto $y = 3$.

Möbius: sketch question; the five interpretations above are the marks.

Trap. Condition d says ∞ , not $-\infty$ and not "does not exist", so **both** sides must go up. Drawing a sign change at $x = 1$ contradicts the condition.

Q5.3 - Interpret six conditions, then sketch

Question. Give the graphical interpretation of each, then sketch $f(x)$.

- a. $f(0) = -3$
- b. $\lim_{x \rightarrow \infty} f(x) = -2$
- c. $\lim_{x \rightarrow 3} f(x) = \infty$
- d. $f'(x) < 0$ on $[3, \infty)$
- e. $f'(x) < 0$ on $(-\infty, -2)$
- f. $f''(x) > 0$ on $[4, \infty)$

Interpretation a. The curve passes through $(0, -3)$.

Interpretation b. Far to the right the curve levels off at height -2 : a horizontal asymptote $y = -2$.

Interpretation c. At $x = 3$ the outputs blow up to $+\infty$ from both sides: a vertical asymptote $x = 3$, upward on both sides.

Interpretation d. On $[3, \infty)$ the function is **decreasing**. Combined with c and b, the branch to the right of $x = 3$ starts at $+\infty$ and falls, levelling onto $y = -2$.

Interpretation e. On $(-\infty, -2)$ the function is also **decreasing**, so on the far left the curve falls as you read left to right.

Interpretation f. On $[4, \infty)$ the curve is **concave up**, holding water. This is consistent with d: a decreasing, concave-up branch flattens onto its asymptote from above, which is exactly how it must meet $y = -2$.

Step: check the conditions are consistent. Right branch: decreasing from $+\infty$ (d), concave up from $x = 4$ on (f), approaching $y = -2$ (b). These agree. Nothing is said about f' on $(-2, 3)$, so you are free there, which is where you route the curve through $(0, -3)$ (a) and up to the asymptote at $x = 3$ (c).

Answer: the checklist your sketch must show.

through $(0, -3)$; dashed $y = -2$; dashed $x = 3$ with $+\infty$ both sides

falling on $(-\infty, -2)$ and on $[3, \infty)$; concave up on $[4, \infty)$

A workable shape: on the far left the curve falls toward a local minimum near $x = -2$, rises through $(0, -3)$ and continues up to $+\infty$ as $x \rightarrow 3^-$; on the right of $x = 3$ it comes down from $+\infty$, decreasing throughout, and flattens onto $y = -2$ from above, curving upward from $x = 4$ on.

Möbius: sketch question; the six interpretations above are the marks.

Trap. Approaching $y = -2$ from *below* on the right. That branch is decreasing and concave up, so it must sit **above** -2 and settle down onto it.

6. Rates of change - interpretation

Q6.1 - Average and instantaneous rate of population change

Question. The population P of a city in year y is $P(y)$. Write a sentence in layman's terms explaining the meaning of:

- a. $\frac{P(15) - P(2)}{13} = 1431$
- b. $P'(10) = 14000$

Part a, step 1: recognise the structure. This is $\frac{\Delta P}{\Delta y}$, a change in population divided by a change in years. The denominator 13 is exactly $15 - 2$, so this is a slope between two points on the population curve.

Part a, step 2: name it. A slope between two separated points is the **average rate of change**, the slope of the secant line.

Part a, step 3: attach units. Population divided by years gives people per year.

Answer a. Between year 2 and year 15, the city's population grew by an average of 1,431 people per year. That is the average over the whole 13-year stretch, so it says nothing about any single year; the population might have surged in some years and shrunk in others and still average this.

Möbius: average rate of change = 1431 people per year over $[2, 15]$

Part b, step 1: recognise the structure. The prime is a derivative evaluated at one instant, not a difference over an interval.

Part b, step 2: name it. This is the **instantaneous rate of change** at year 10, the slope of the tangent line to the population curve there.

Answer b. In year 10, the city's population was growing at a rate of 14,000 people per year at that instant. It is the growth rate at that moment, so if the city kept growing at exactly that pace for a full year it would add about 14,000 people, but the actual increase over that year may differ because the rate itself keeps changing.

Möbius: instantaneous rate of change = 14000 people per year at $y=10$

Trap. Saying "the population increased by 14,000 in year 10". A derivative is a rate, not a total. Part a is an actual average over 13 years; part b is a speed at a single instant.

7. The derivative from its definition

Q7.1 - Derivative of cotangent from the definition

Question. Use the definition of the derivative as a limit to compute the derivative of $\cot x$.

Step 1: write the definition. The word "definition" means the limit must appear; a table lookup earns zero.

$$\frac{d}{dx} \cot x = \lim_{h \rightarrow 0} \frac{\cot(x+h) - \cot x}{h}$$

Step 2: convert to sine and cosine. Cotangent addition formulas are ugly; the sine and cosine form is far cleaner.

$$\cot(x+h) - \cot x = \frac{\cos(x+h)}{\sin(x+h)} - \frac{\cos x}{\sin x}$$

Step 3: combine over a common denominator.

$$= \frac{\cos(x+h) \sin x - \cos x \sin(x+h)}{\sin(x+h) \sin x}$$

Step 4: recognise the sine difference identity. The numerator is exactly $\sin(A-B) = \sin A \cos B - \cos A \sin B$ with $A = x$ and $B = x+h$.

$$\cos(x+h) \sin x - \cos x \sin(x+h) = \sin(x - (x+h)) = \sin(-h) = -\sin h$$

This collapse is the whole point of the question.

Step 5: put it back into the limit.

$$\frac{d}{dx} \cot x = \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{-\sin h}{\sin(x+h) \sin x}$$

Step 6: separate the special limit.

$$= \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \right) \cdot \frac{1}{\sin(x+h) \sin x}$$

Step 7: evaluate each factor. The first factor tends to -1 by the special limit $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$. The second tends to $\frac{1}{\sin x \sin x}$ because sine is continuous.

$$= -1 \cdot \frac{1}{\sin^2 x} = -\frac{1}{\sin^2 x}$$

Step 8: convert to the standard form. Since $\csc x = \frac{1}{\sin x}$:

Answer.

$$\frac{d}{dx} \cot x = -\frac{1}{\sin^2 x} = -\csc^2 x$$

Möbius: $-\csc(x)^2$ or equivalently $-1/\sin(x)^2$

Trap. Writing $\sin(x - (x+h)) = \sin(h)$ and losing the minus sign, which turns the answer into $+\csc^2 x$. The argument is $-h$, and sine is odd.

8. Derivative rules and computation

Q8.1 - Four derivatives, power chain and quotient rules

Question. Compute the derivatives. Simplification is not required.

- a. $y = 4x^5 + 3x^4 - 6x^3 + 6$
- b. $y = \frac{2x - 16}{(x + 3)^2}$
- c. $y = \sin(2x^2 - x + 1)$
- d. $y = (-4x^3 - x^2 + 3x + 7)^4$

Part a: power rule term by term. Multiply each coefficient by its exponent and drop the exponent by one; the constant 6 dies.

$$y' = 20x^4 + 12x^3 - 18x^2$$

Answer a. $y' = 20x^4 + 12x^3 - 18x^2$. **Möbius:** $20 \cdot x^4 + 12 \cdot x^3 - 18 \cdot x^2$

Part b, step 1: set up the quotient rule. Let $f = 2x - 16$ and $g = (x + 3)^2$, so $f' = 2$ and $g' = 2(x + 3)$ by the chain rule.

$$y' = \frac{f'g - fg'}{g^2} = \frac{2(x + 3)^2 - (2x - 16) \cdot 2(x + 3)}{(x + 3)^4}$$

Part b, step 2: cancel one factor of $(x + 3)$. Every term on top carries an $(x + 3)$, so one cancels against the bottom.

$$= \frac{2(x + 3) - 2(2x - 16)}{(x + 3)^3}$$

Part b, step 3: expand the numerator.

$$= \frac{2x + 6 - 4x + 32}{(x + 3)^3} = \frac{38 - 2x}{(x + 3)^3}$$

Answer b.

$$y' = \frac{38 - 2x}{(x + 3)^3}$$

Möbius: $(38 - 2 \cdot x) / (x + 3)^3$

Part c: chain rule. Outside is $\sin(\)$, whose derivative is $\cos(\)$; inside is $2x^2 - x + 1$, whose derivative is $4x - 1$.

$$y' = \cos(2x^2 - x + 1) \cdot (4x - 1)$$

Answer c. Möbius: $\cos(2 \cdot x^2 - x + 1) \cdot (4 \cdot x - 1)$

Part d: chain rule with the power rule outside. Outside is $(\)^4$, giving $4(\)^3$; inside derivative is $-12x^2 - 2x + 3$.

$$y' = 4(-4x^3 - x^2 + 3x + 7)^3 (-12x^2 - 2x + 3)$$

Answer d. Möbius: $4 \cdot (-4 \cdot x^3 - x^2 + 3 \cdot x + 7)^3 \cdot (-12 \cdot x^2 - 2 \cdot x + 3)$

Q8.2 - Four derivatives, including a second derivative**Question.** Find each derivative, showing work. Simplification is not required.

- a. $\frac{d^2}{dx^2} \cot(2x)$
- b. $\frac{d}{dx} \sec(x^2 - 3x)$
- c. $\frac{d}{dx} \frac{x^2 - \cos(3x)}{x \sin(2x)}$
- d. $\left. \frac{d}{dx} \frac{f(x)}{g(x)} \right|_{x=1}$ where $f(1) = 3$, $g(1) = 6$, $f'(x) = \sqrt{x}$, $g'(x) = 3 \sin\left(\frac{\pi x}{3}\right)$

Part a, step 1: first derivative. $\frac{d}{dx} \cot u = -\csc^2 u$ with $u = 2x$, so multiply by $u' = 2$.

$$\frac{d}{dx} \cot(2x) = -2 \csc^2(2x)$$

Part a, step 2: differentiate again. Write $\csc^2(2x)$ as $(\csc(2x))^2$ and use the chain rule twice.

$$\frac{d}{dx} (\csc(2x))^2 = 2 \csc(2x) \cdot \frac{d}{dx} \csc(2x)$$

$$\frac{d}{dx} \csc(2x) = -\csc(2x) \cot(2x) \cdot 2$$

Part a, step 3: combine.

$$\frac{d}{dx} \csc^2(2x) = 2 \csc(2x) \cdot (-2 \csc(2x) \cot(2x)) = -4 \csc^2(2x) \cot(2x)$$

Part a, step 4: apply the leading -2.

$$\frac{d^2}{dx^2} \cot(2x) = -2 \cdot (-4 \csc^2(2x) \cot(2x)) = 8 \csc^2(2x) \cot(2x)$$

Answer a.

$$\frac{d^2}{dx^2} \cot(2x) = 8 \csc^2(2x) \cot(2x)$$

Möbius: $8 \csc(2x)^2 \cot(2x)$ **Trap.** Two minus signs and two inner 2s appear here; the final answer is **positive**. Losing one factor of 2 gives 4 instead of 8.**Part b: chain rule with the secant rule.** $\frac{d}{dx} \sec u = \sec u \tan u$, with $u = x^2 - 3x$ and $u' = 2x - 3$.

$$\frac{d}{dx} \sec(x^2 - 3x) = \sec(x^2 - 3x) \tan(x^2 - 3x) (2x - 3)$$

Answer b. Möbius: $\sec(x^2 - 3x) \tan(x^2 - 3x) (2x - 3)$ **Part c, step 1: name the pieces.**

$$f = x^2 - \cos(3x), \quad g = x \sin(2x)$$

Part c, step 2: differentiate each. For f , the derivative of $-\cos(3x)$ is $+3 \sin(3x)$. For g , use the product rule on x times $\sin(2x)$.

$$f' = 2x + 3 \sin(3x), \quad g' = \sin(2x) + 2x \cos(2x)$$

Part c, step 3: assemble the quotient rule.

$$\frac{d}{dx} \frac{f}{g} = \frac{(2x + 3 \sin(3x))(x \sin(2x)) - (x^2 - \cos(3x))(\sin(2x) + 2x \cos(2x))}{(x \sin(2x))^2}$$

Answer c. As above; no simplification required.

Möbius: $((2x+3\sin(3x))*(x\sin(2x)) - (x^2-\cos(3x))*(\sin(2x)+2x\cos(2x)))/(x\sin(2x))^2$

Part d, step 1: write the quotient rule at a point.

$$\left(\frac{f}{g}\right)' \bigg|_{x=1} = \frac{f'(1)g(1) - f(1)g'(1)}{(g(1))^2}$$

Part d, step 2: evaluate the two derivatives at $x = 1$.

$$f'(1) = \sqrt{1} = 1$$

$$g'(1) = 3 \sin\left(\frac{\pi}{3}\right) = 3 \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

Part d, step 3: substitute all four numbers.

$$= \frac{(1)(6) - (3)\left(\frac{3\sqrt{3}}{2}\right)}{6^2} = \frac{6 - \frac{9\sqrt{3}}{2}}{36}$$

Part d, step 4: clear the inner fraction. Multiply numerator and denominator by 2.

$$= \frac{12 - 9\sqrt{3}}{72} = \frac{4 - 3\sqrt{3}}{24}$$

Answer d.

$$\frac{d}{dx} \frac{f}{g} \bigg|_{x=1} = \frac{4 - 3\sqrt{3}}{24} \approx -0.04984$$

Möbius: $(4-3*\text{sqrt}(3))/24$

Trap. $\sin\left(\frac{\pi x}{3}\right)$ at $x = 1$ is $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$, in radians. Reading it as $\sin(1^\circ)$ or forgetting the outer 3 both break the answer.

Q8.3 - Five derivatives, nested chain and quotient rules**Question.** Compute the derivatives. Simplification is not required.

- a. $y = \sin x \cdot \cos(\sin x^2)$
- b. $y = \left(\frac{1+x^3}{1-x^2}\right)^{1/3}$
- c. $y = \sqrt{1 + \sqrt{1+x}}$
- d. $y = \frac{\sqrt{x^2-1}}{x^2-2x-8}$
- e. $y = \sec^2\left(\frac{x+1}{x-2}\right)$

Part a, step 1: identify the outer structure. This is a **product** of $\sin x$ and $\cos(\sin x^2)$, so the product rule runs first.**Part a, step 2: differentiate the second factor, which is three layers deep.** Layers: $\cos(\)$, then $\sin(\)$, then x^2 .

$$\frac{d}{dx} \cos(\sin x^2) = -\sin(\sin x^2) \cdot \cos(x^2) \cdot 2x$$

Part a, step 3: assemble the product rule.

$$y' = \cos x \cos(\sin x^2) - \sin x \sin(\sin x^2) \cos(x^2) (2x)$$

Answer a. Möbius: $\cos(x) \cdot \cos(\sin(x^2)) - \sin(x) \cdot \sin(\sin(x^2)) \cdot \cos(x^2) \cdot 2x$ **Trap.** $\sin x^2$ means $\sin(x^2)$, not $(\sin x)^2$. The inner derivative is $\cos(x^2) \cdot 2x$.**Part b, step 1: outer power rule.** Let $u = \frac{1+x^3}{1-x^2}$, so $y = u^{1/3}$ and $y' = \frac{1}{3} u^{-2/3} u'$.**Part b, step 2: quotient rule on u .**

$$u' = \frac{3x^2(1-x^2) - (1+x^3)(-2x)}{(1-x^2)^2}$$

Part b, step 3: expand the numerator.

$$3x^2 - 3x^4 + 2x + 2x^4 = -x^4 + 3x^2 + 2x$$

Part b, step 4: assemble.

$$y' = \frac{1}{3} \left(\frac{1+x^3}{1-x^2}\right)^{-2/3} \cdot \frac{-x^4 + 3x^2 + 2x}{(1-x^2)^2}$$

Answer b. Möbius: $(1/3) * ((1+x^3)/(1-x^2))^{(-2/3)} * (-x^4+3x^2+2x)/(1-x^2)^2$ **Part c, step 1: peel the outer root.** With $y = \sqrt{v}$ and $v = 1 + \sqrt{1+x}$:

$$y' = \frac{1}{2\sqrt{1+\sqrt{1+x}}} \cdot v'$$

Part c, step 2: differentiate the inner root. $\frac{d}{dx}\sqrt{1+x} = \frac{1}{2\sqrt{1+x}}$, and the derivative of the constant 1 is zero.

Part c, step 3: multiply.

$$y' = \frac{1}{2\sqrt{1+\sqrt{1+x}}} \cdot \frac{1}{2\sqrt{1+x}} = \frac{1}{4\sqrt{1+x}\sqrt{1+\sqrt{1+x}}}$$

Answer c.

$$y' = \frac{1}{4\sqrt{1+x}\sqrt{1+\sqrt{1+x}}}$$

Möbius: $1/(4*\text{sqrt}(1+x)*\text{sqrt}(1+\text{sqrt}(1+x)))$

Part d, step 1: quotient rule with a chain rule inside. Let $f = \sqrt{x^2-1}$ and $g = x^2-2x-8$.

$$f' = \frac{2x}{2\sqrt{x^2-1}} = \frac{x}{\sqrt{x^2-1}}, \quad g' = 2x-2$$

Part d, step 2: assemble.

$$y' = \frac{\frac{x}{\sqrt{x^2-1}}(x^2-2x-8) - \sqrt{x^2-1}(2x-2)}{(x^2-2x-8)^2}$$

Answer d. Möbius: $((x/\text{sqrt}(x^2-1))*(x^2-2*x-8) - \text{sqrt}(x^2-1)*(2*x-2))/(x^2-2*x-8)^2$

Part e, step 1: read the layers. $y = (\sec u)^2$ with $u = \frac{x+1}{x-2}$: a square, then a secant, then a quotient.

Part e, step 2: differentiate the inner quotient.

$$u' = \frac{(1)(x-2) - (x+1)(1)}{(x-2)^2} = \frac{x-2-x-1}{(x-2)^2} = \frac{-3}{(x-2)^2}$$

Part e, step 3: outer layers.

$$\frac{d}{dx}(\sec u)^2 = 2 \sec u \cdot \sec u \tan u \cdot u' = 2 \sec^2 u \tan u \cdot u'$$

Part e, step 4: substitute.

$$y' = 2 \sec^2\left(\frac{x+1}{x-2}\right) \tan\left(\frac{x+1}{x-2}\right) \cdot \frac{-3}{(x-2)^2} = \frac{-6 \sec^2\left(\frac{x+1}{x-2}\right) \tan\left(\frac{x+1}{x-2}\right)}{(x-2)^2}$$

Answer e. Möbius: $-6*\text{sec}((x+1)/(x-2))^2*\tan((x+1)/(x-2))/(x-2)^2$

Q8.4 - Two derivatives, naming the rules used

Question. Give the derivatives and **state which rules** you apply. Simplification is not required.

- a. $\frac{d}{dx} \frac{\tan(2x)}{\sqrt{x}}$
- b. $\frac{d}{dx} \cos^3(x^2) \Big|_{x=\sqrt{\pi}/2}$

Part a, step 1: name the rules. **Quotient rule** on the outside, **chain rule** on $\tan(2x)$, **power rule** on $\sqrt{x} = x^{1/2}$.

Part a, step 2: the pieces.

$$f = \tan(2x), \quad f' = 2 \sec^2(2x), \quad g = x^{1/2}, \quad g' = \frac{1}{2\sqrt{x}}$$

Part a, step 3: assemble, noting $g^2 = x$.

$$\frac{d}{dx} \frac{\tan(2x)}{\sqrt{x}} = \frac{2 \sec^2(2x) \sqrt{x} - \tan(2x) \cdot \frac{1}{2\sqrt{x}}}{x}$$

Answer a. Rules used: quotient, chain, power.

Möbius: $(2*\sec(2*x)^2*\sqrt{x} - \tan(2*x)/(2*\sqrt{x}))/x$

Part b, step 1: name the rules. Chain rule twice: the outer cube, then the cosine, then the inner x^2 . Read the notation first: $\cos^3(x^2)$ means $[\cos(x^2)]^3$.

Part b, step 2: differentiate.

$$\begin{aligned} \frac{d}{dx} [\cos(x^2)]^3 &= 3 [\cos(x^2)]^2 \cdot (-\sin(x^2)) \cdot 2x \\ &= -6x \cos^2(x^2) \sin(x^2) \end{aligned}$$

Part b, step 3: evaluate at $x = \frac{\sqrt{\pi}}{2}$. First find the inner value, which is the move that makes this clean.

$$x^2 = \left(\frac{\sqrt{\pi}}{2}\right)^2 = \frac{\pi}{4}$$

Part b, step 4: use the exact values at $\frac{\pi}{4}$.

$$\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2} \implies \cos^2 \frac{\pi}{4} = \frac{1}{2}, \quad \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

Part b, step 5: substitute.

$$-6 \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = -\frac{6\sqrt{2}\sqrt{\pi}}{8} = -\frac{3\sqrt{2\pi}}{4}$$

Answer b.

$$\left. \frac{d}{dx} \cos^3(x^2) \right|_{x=\sqrt{\pi}/2} = -\frac{3\sqrt{2\pi}}{4} \approx -1.87997$$

Rules used: chain rule, twice.

Möbius: $-3*\text{sqrt}(2*\text{pi})/4$

9. Tangent lines

Q9.1 - Tangent line perpendicular to a given line

Question. Find the values of x where the tangent line to $f(x) = 3x^2 + 4x - 3$ is perpendicular to the line $y = 6x + 2$.

Step 1: read the slope of the given line. In $y = mx + b$ form, $m = 6$.

Step 2: convert perpendicular into a number. Perpendicular slopes are negative reciprocals, so the tangent must have slope

$$m_{\perp} = -\frac{1}{6}$$

Step 3: the tangent slope is the derivative.

$$f'(x) = 6x + 4$$

Step 4: set the derivative equal to the required slope. This is the equation the question is really asking you to solve.

$$6x + 4 = -\frac{1}{6}$$

Step 5: solve.

$$6x = -\frac{1}{6} - 4 = -\frac{1}{6} - \frac{24}{6} = -\frac{25}{6}$$

$$x = -\frac{25}{36}$$

Answer.

$$x = -\frac{25}{36} \approx -0.6944$$

Möbius: $-25/36$

Check. $f'\left(-\frac{25}{36}\right) = 6\left(-\frac{25}{36}\right) + 4 = -\frac{25}{6} + \frac{24}{6} = -\frac{1}{6}$, and $6 \times \left(-\frac{1}{6}\right) = -1$, which confirms perpendicularity.

Trap. Setting $f'(x) = 6$, which finds where the tangent is **parallel** to the line, not perpendicular. The question asks for the negative reciprocal.

10. Implicit differentiation

Q10.1 - Find the derivative implicitly

Question. Find y' by implicit differentiation:

$$x^3y + xy^2 = 4xy + 7$$

Step 1: differentiate both sides with respect to x . Every term with a y in it needs the product rule, the chain rule, or both.

Step 2: term by term.

$$\frac{d}{dx}(x^3y) = 3x^2y + x^3y' \quad (\text{product rule})$$

$$\frac{d}{dx}(xy^2) = y^2 + x \cdot 2yy' = y^2 + 2xyy' \quad (\text{product then chain})$$

$$\frac{d}{dx}(4xy) = 4y + 4xy' \quad (\text{product rule})$$

$$\frac{d}{dx}(7) = 0$$

Step 3: write the differentiated equation.

$$3x^2y + x^3y' + y^2 + 2xyy' = 4y + 4xy'$$

Step 4: move every y' term to the left and everything else to the right.

$$x^3y' + 2xyy' - 4xy' = 4y - 3x^2y - y^2$$

Step 5: factor out y' .

$$y'(x^3 + 2xy - 4x) = 4y - 3x^2y - y^2$$

Step 6: divide.

Answer.

$$y' = \frac{4y - 3x^2y - y^2}{x^3 + 2xy - 4x}$$

Equivalently, factoring y out of the top and x out of the bottom:

$$y' = \frac{y(4 - 3x^2 - y)}{x(x^2 + 2y - 4)}$$

Möbius: $(4y - 3x^2y - y^2)/(x^3 + 2xy - 4x)$

Trap. Differentiating xy^2 as $2xyy'$ and forgetting the y^2 from the product rule. Both the "derivative of x " term and the "derivative of y^2 " term must appear.

Q10.2 - Tangent line to an implicit curve

Question. Find the equation of the tangent line to

$$y^3 + yx^2 + x^2 = 3y^2$$

at the point $(1, 1)$.

Step 1: confirm the point is on the curve. Always do this first; if it is not on the curve, the question has been misread.

$$1^3 + (1)(1)^2 + 1^2 = 1 + 1 + 1 = 3, \quad 3(1)^2 = 3$$

Both sides equal 3, so the point does lie on the curve.

Step 2: differentiate both sides implicitly.

$$\frac{d}{dx}(y^3) = 3y^2y'$$

$$\frac{d}{dx}(yx^2) = y'x^2 + 2xy \quad (\text{product rule})$$

$$\frac{d}{dx}(x^2) = 2x, \quad \frac{d}{dx}(3y^2) = 6yy'$$

Step 3: write the differentiated equation.

$$3y^2y' + x^2y' + 2xy + 2x = 6yy'$$

Step 4: substitute the point now, rather than solving symbolically. Because only one point is needed, plugging in $x = 1$, $y = 1$ immediately is far faster and less error-prone.

$$3(1)y' + (1)y' + 2(1)(1) + 2(1) = 6(1)y'$$

$$3y' + y' + 2 + 2 = 6y'$$

Step 5: solve for y' .

$$4y' + 4 = 6y' \implies 4 = 2y' \implies y' = 2$$

Step 6: build the tangent line with point-slope form.

$$y - 1 = 2(x - 1)$$

Answer.

$$y = 2x - 1$$

Möbius: $2x - 1$, slope 2 at $(1, 1)$

Trap. Solving for y' in full symbolic generality and then substituting. It works, but it is slower and every extra algebra step is a chance to lose a term.

11. Related rates

Q11.1 - Gravel cone growing in height

Question. Gravel is dumped at $\frac{dV}{dt} = 0.5 \text{ m}^3/\text{min}$. It forms a cone whose base diameter and height are always equal. Find $\frac{dh}{dt}$ when $h = 4 \text{ m}$.

Step 1: write down what is given and what is wanted, as rates.

$$\text{Given } \frac{dV}{dt} = 0.5, \quad \text{Want } \frac{dh}{dt} \text{ when } h = 4$$

Step 2: write the geometry.

$$V = \frac{1}{3}\pi r^2 h$$

Step 3: use the constraint to eliminate r . "Base diameter equals height" means $2r = h$, so $r = \frac{h}{2}$. This must be done **before** differentiating, because there is no given rate for r .

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{1}{3}\pi \cdot \frac{h^2}{4} \cdot h = \frac{\pi h^3}{12}$$

Step 4: differentiate with respect to t .

$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \frac{dh}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}$$

Step 5: substitute the numbers, only now.

$$0.5 = \frac{\pi(4)^2}{4} \frac{dh}{dt} = 4\pi \frac{dh}{dt}$$

Step 6: solve.

$$\frac{dh}{dt} = \frac{0.5}{4\pi} = \frac{1}{8\pi}$$

Answer.

$$\frac{dh}{dt} = \frac{1}{8\pi} \approx 0.0398 \text{ m/min}$$

Möbius: $1/(8\pi)$

Trap. Reading "diameter equals height" as $r = h$. The diameter is $2r$, so $r = \frac{h}{2}$; using $r = h$ makes the answer four times too small.

Q11.2 - Rocket tracked by a radar station

Question. A rocket rises vertically, tracked by a radar station 5 mi from the pad. Find $\frac{dy}{dt}$ when $y = 4$ mi and $\frac{dz}{dt} = 2000$ mi/h, where z is the rocket's distance from the station.

Step 1: name the variables and draw the triangle. The horizontal leg is fixed at 5; the vertical leg y is the rocket's height; the hypotenuse z is the radar distance.

$$\text{Given } \frac{dz}{dt} = 2000, \quad \text{Want } \frac{dy}{dt} \text{ when } y = 4$$

Step 2: write the relationship. The 5 is a constant, not a variable, which is what makes this solvable.

$$z^2 = 5^2 + y^2 = 25 + y^2$$

Step 3: find the missing side at that instant.

$$z = \sqrt{25 + 16} = \sqrt{41}$$

Step 4: differentiate with respect to t .

$$2z \frac{dz}{dt} = 2y \frac{dy}{dt}$$

$$z \frac{dz}{dt} = y \frac{dy}{dt}$$

Step 5: substitute and solve.

$$\sqrt{41} (2000) = 4 \frac{dy}{dt} \implies \frac{dy}{dt} = \frac{2000\sqrt{41}}{4} = 500\sqrt{41}$$

Answer.

$$\frac{dy}{dt} = 500\sqrt{41} \approx 3201.6 \text{ mi/h}$$

Möbius: $500 \cdot \sqrt{41}$

Sanity check. The rocket must rise faster than the radar distance grows, because only part of its motion is along the line of sight. $3201.6 > 2000$, as required.

Trap. Using $z = 4$ and $y = \sqrt{41}$, that is, swapping the leg and the hypotenuse. The hypotenuse is always the largest side.

12. Linear approximation and differentials

Q12.1 - Approximate a square root with differentials

Question. Use differentials to approximate $\sqrt{9.2}$.

Step 1: choose the function and the base point. Pick a as the nearest value whose root you know exactly.

$$f(x) = \sqrt{x}, \quad a = 9, \quad dx = 0.2$$

Step 2: write the differential formula.

$$f(a + dx) \approx f(a) + f'(a) dx$$

Step 3: compute the derivative at the base point.

$$f'(x) = \frac{1}{2\sqrt{x}} \implies f'(9) = \frac{1}{2 \cdot 3} = \frac{1}{6}$$

Step 4: assemble.

$$\sqrt{9.2} \approx 3 + \frac{1}{6}(0.2) = 3 + \frac{0.2}{6} = 3 + \frac{1}{30}$$

Answer.

$$\sqrt{9.2} \approx 3 + \frac{1}{30} = \frac{91}{30} \approx 3.0333$$

The true value is 3.03315, so the estimate is high by about 0.0002, which is expected: $\sqrt{x}$ is concave down, so its tangent line lies above the curve.

Möbius: 91/30 or 3.0333

Q12.2 - Estimate a sine with linearization

Question. Use linearization to estimate $\sin(62^\circ)$.

Step 1: choose the base point. 60° is the nearest angle with exact values.

$$f(x) = \sin x, \quad a = 60^\circ = \frac{\pi}{3}$$

Step 2: convert the step to radians. This is the step that decides the whole answer; calculus trig formulas are only valid in radians.

$$dx = 2^\circ = 2 \cdot \frac{\pi}{180} = \frac{\pi}{90} \approx 0.0349066$$

Step 3: write the linearization.

$$\sin(a + dx) \approx \sin a + \cos a dx$$

Step 4: substitute the exact values.

$$\sin(62^\circ) \approx \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{\pi}{90}$$

Step 5: evaluate.

$$\approx 0.8660254 + 0.0174533 = 0.8834787$$

Answer.

$$\sin(62^\circ) \approx \frac{\sqrt{3}}{2} + \frac{\pi}{180} \approx 0.8835$$

The true value is 0.8829476, so the estimate is high by about 0.0005, again because sine is concave down on this stretch.

Möbius: $\sqrt{3}/2 + \pi/180$ or 0.8835

Trap. Using $dx = 2$ instead of $dx = \frac{\pi}{90}$. That gives ≈ 1.87 , which is impossible for a sine.

Q12.3 - Two estimates to four decimal places

Question. Use differentials to estimate to four decimal places, showing the process.

- a. $\cos(62^\circ)$
- b. $\sqrt{16.4}$

Part a, step 1: base point and step.

$$f(x) = \cos x, \quad a = \frac{\pi}{3}, \quad dx = \frac{\pi}{90}$$

Part a, step 2: the derivative carries a minus sign.

$$f'(x) = -\sin x \implies f'\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

Part a, step 3: assemble.

$$\cos(62^\circ) \approx \frac{1}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\pi}{90}$$

Part a, step 4: evaluate.

$$\approx 0.5 - 0.8660254(0.0349066) = 0.5 - 0.0302300 = 0.4697700$$

Answer a.

$$\cos(62^\circ) \approx 0.4698$$

The true value is 0.4694716. Note the estimate is now **high**, and the sign of the error flipped versus Q12.2 because cosine is decreasing here.

Möbius: $1/2 - \sqrt{3} \cdot \pi/180$ or 0.4698

Part b, step 1: base point and step.

$$f(x) = \sqrt{x}, \quad a = 16, \quad dx = 0.4$$

Part b, step 2: derivative at the base point.

$$f'(16) = \frac{1}{2\sqrt{16}} = \frac{1}{8}$$

Part b, step 3: assemble and evaluate.

$$\sqrt{16.4} \approx 4 + \frac{1}{8}(0.4) = 4 + 0.05 = 4.05$$

Answer b.

$$\sqrt{16.4} \approx 4.0500$$

The true value is 4.0496913, so the estimate is high by about 0.0003.

Möbius: 4.05 or 81/20

Trap. The question says four decimal places, so write 4.0500, not 4.05. Trailing zeros are the requested precision.

13. Newton's method

Q13.1 - Approximate a root with Newton's method

Question. Use Newton's method to approximate the solution of

$$x^4 + 2x - 5 = 0 \quad \text{in } [1, 2]$$

Step 1: confirm a root exists in the interval. Check the sign change, which is the Intermediate Value Theorem justification.

$$f(1) = 1 + 2 - 5 = -2, \quad f(2) = 16 + 4 - 5 = 15$$

The sign changes from negative to positive, and f is continuous, so a root lies in $(1, 2)$.

Step 2: write the derivative and the iteration formula.

$$f'(x) = 4x^3 + 2$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^4 + 2x_n - 5}{4x_n^3 + 2}$$

Step 3: choose a starting value. Take $x_0 = 1$, an endpoint of the given interval.

Step 4: iterate, keeping all decimals.

$$x_1 = 1 - \frac{-2}{6} = 1 + \frac{1}{3} = 1.333333333$$

$$x_2 = 1.333333333 - \frac{f(1.333333333)}{f'(1.333333333)} = 1.261290323$$

$$x_3 = 1.255964749$$

$$x_4 = 1.255937549$$

$$x_5 = 1.255937548$$

Step 5: stop when the digits stop moving. x_4 and x_5 agree to eight decimal places, so the method has converged.

Answer.

$$x \approx 1.2559$$

To more places, $x \approx 1.255937548$.

Möbius: 1.2559

Check. $f(1.255937548) \approx 0$ to eleven decimal places, and 1.2559 lies inside $[1, 2]$ as required.

Trap. Rounding to four decimals *between* iterations. Newton's method converges by doubling correct digits each step, and rounding early throws that away. Keep the full display and round only the final answer.

14. Curve sketching with calculus

Q14.1 - Sketch two curves with a full calculus analysis

Question. Sketch each graph, stating all critical points, cusps, vertical asymptotes and points of inflection.

- a. $f(x) = \frac{x^2}{x^2 - 1}$
- b. $f(x) = x + \sin x$

Part a, step 1: domain and vertical asymptotes.

$$x^2 - 1 = 0 \implies x = \pm 1$$

Neither factor cancels, so both are genuine vertical asymptotes.

$$\text{Domain: } x \neq \pm 1$$

Part a, step 2: horizontal asymptote. Equal degrees, so take the ratio of leading coefficients.

$$\lim_{x \rightarrow \pm\infty} \frac{x^2}{x^2 - 1} = 1 \implies y = 1$$

Part a, step 3: first derivative, by the quotient rule.

$$f'(x) = \frac{2x(x^2 - 1) - x^2(2x)}{(x^2 - 1)^2} = \frac{2x^3 - 2x - 2x^3}{(x^2 - 1)^2} = \frac{-2x}{(x^2 - 1)^2}$$

Part a, step 4: critical points. $f'(x) = 0$ requires $-2x = 0$, so $x = 0$. f' is undefined at $x = \pm 1$, but f is undefined there too, so those are asymptotes, not critical points.

$$f(0) = \frac{0}{-1} = 0 \implies \text{critical point } (0, 0)$$

Part a, step 5: classify it. The denominator $(x^2 - 1)^2$ is always positive, so the sign of f' is the sign of $-2x$.

$$f' > 0 \text{ for } x < 0, \quad f' < 0 \text{ for } x > 0$$

Increasing then decreasing, so $(0, 0)$ is a **local maximum**.

Part a, step 6: second derivative.

$$f''(x) = \frac{6x^2 + 2}{(x^2 - 1)^3}$$

Part a, step 7: inflection points. An inflection needs $f'' = 0$ with a sign change.

$$6x^2 + 2 = 0 \implies x^2 = -\frac{1}{3}$$

This has **no real solution**, so the numerator never vanishes.

There are no points of inflection.

Concavity is decided entirely by the denominator: $f'' > 0$ when $|x| > 1$ (concave up) and $f'' < 0$ when $|x| < 1$ (concave down). The sign does change across $x = \pm 1$, but those are not inflection points because f is not defined there.

Answer a.

Vertical asymptotes $x = -1, x = 1$; horizontal asymptote $y = 1$

Critical point $(0, 0)$, a local maximum; no cusps; no inflection points

Shape: three branches. The outer two sit above $y = 1$ and fall toward it as $|x|$ grows; the middle branch is a downward cap through $(0, 0)$, dropping to $-\infty$ at both $x = -1^+$ and $x = 1^-$.

Möbius: VA $x = -1, x = 1$; HA $y = 1$; critical point $(0, 0)$ local max; inflection points: none

Part b, step 1: domain. $x + \sin x$ is defined for all real x ; no asymptotes, no cusps.

Part b, step 2: first derivative.

$$f'(x) = 1 + \cos x$$

Part b, step 3: critical points.

$$1 + \cos x = 0 \implies \cos x = -1 \implies x = \pi + 2k\pi, \quad k \in \mathbb{Z}$$

Part b, step 4: classify them. Since $\cos x \geq -1$ always, $f'(x) = 1 + \cos x \geq 0$ **everywhere**. So f is increasing on all of $\mathbb{R}$ and never turns around.

These are horizontal tangents, not maxima or minima.

At $x = \pi$, $f(\pi) = \pi + 0 = \pi$, so (π, π) is a critical point where the curve momentarily flattens and then keeps climbing.

Part b, step 5: second derivative and inflection points.

$$f''(x) = -\sin x$$

$$-\sin x = 0 \implies x = k\pi, \quad k \in \mathbb{Z}$$

The sign of $-\sin x$ genuinely alternates across each $k\pi$, so **every** $x = k\pi$ is an inflection point.

$$f(k\pi) = k\pi + \sin(k\pi) = k\pi \implies \text{inflection points } (k\pi, k\pi)$$

Every inflection point lies exactly on the line $y = x$.

Part b, step 6: concavity. $f'' = -\sin x > 0$ where $\sin x < 0$, that is on $(\pi, 2\pi)$ and its 2π -translates: concave up. Concave down on $(0, \pi)$ and its translates.

Answer b.

No asymptotes, no cusps, no maxima or minima

Critical points at $x = \pi + 2k\pi$ (horizontal tangents on an increasing curve)

Inflection points at $(k\pi, k\pi)$ for every integer k

Shape: a staircase-like curve rising forever, weaving around the line $y = x$, touching it at every multiple of π , with a flat spot at $x = \pi, 3\pi, 5\pi, \dots$

Möbius: critical points $x = \pi + 2k\pi$; inflection points $(k\pi, k\pi)$; no asymptotes; increasing everywhere

Trap. Calling $x = \pi$ a local maximum because $f'(\pi) = 0$. The first derivative does not change sign there, so it is neither a max nor a min.

15. Extreme values and optimization

Q15.1 - All maxima and minima on closed intervals

Question. Find all maxima and minima on the indicated intervals.

- a. $f(x) = 2x^{5/3} - 5x^{4/3}$ on $[-1, 20]$
- b. $f(x) = x + \cos x$ on $[-\pi, 2\pi]$

Part a, step 1: differentiate.

$$f'(x) = \frac{10}{3}x^{2/3} - \frac{20}{3}x^{1/3}$$

Part a, step 2: factor. Pull out the common $\frac{10}{3}x^{1/3}$.

$$f'(x) = \frac{10}{3}x^{1/3}(x^{1/3} - 2)$$

Part a, step 3: find the critical numbers.

$$x^{1/3} = 0 \implies x = 0$$

$$x^{1/3} = 2 \implies x = 8$$

Both lie in $[-1, 20]$. Note f' is defined everywhere here, since $x^{1/3}$ and $x^{2/3}$ accept negatives.

Part a, step 4: evaluate at critical numbers and endpoints. For $x = -1$, use the real cube root: $(-1)^{1/3} = -1$, so $(-1)^{5/3} = -1$ and $(-1)^{4/3} = 1$.

$$f(-1) = 2(-1) - 5(1) = -7$$

$$f(0) = 0$$

$$f(8) = 2(32) - 5(16) = 64 - 80 = -16$$

$$f(20) = 2 \cdot 20^{5/3} - 5 \cdot 20^{4/3} \approx 294.72 - 271.44 = 23.28$$

Part a, step 5: compare.

Answer a.

$$\text{Absolute maximum } f(20) = 2 \cdot 20^{5/3} - 5 \cdot 20^{4/3} \approx 23.281 \text{ at } x = 20$$

$$\text{Absolute minimum } f(8) = -16 \text{ at } x = 8$$

Also present: a **local maximum** at $(0, 0)$, since f rises on $[-1, 0]$, falls on $[0, 8]$, then rises again on $[8, 20]$.

Möbius: $\max 2 \cdot 20^{5/3} - 5 \cdot 20^{4/3}$ at $x=20$; $\min -16$ at $x=8$

Trap. Treating $(-1)^{4/3}$ as undefined or negative. Read it as $\left((-1)^{1/3}\right)^4 = (-1)^4 = 1$; the even outer power makes it positive.

Part b, step 1: differentiate.

$$f'(x) = 1 - \sin x$$

Part b, step 2: critical numbers.

$$1 - \sin x = 0 \implies \sin x = 1 \implies x = \frac{\pi}{2} + 2k\pi$$

Within $[-\pi, 2\pi]$ the only such value is $x = \frac{\pi}{2}$.

Part b, step 3: note the monotonicity. $\sin x \leq 1$ always, so $f'(x) = 1 - \sin x \geq 0$ everywhere. The function is **increasing** on the whole interval, touching slope zero only at $x = \frac{\pi}{2}$. This already tells you the extremes must be the endpoints.

Part b, step 4: evaluate.

$$f(-\pi) = -\pi + \cos(-\pi) = -\pi - 1 \approx -4.1416$$

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 0 = \frac{\pi}{2} \approx 1.5708$$

$$f(2\pi) = 2\pi + \cos(2\pi) = 2\pi + 1 \approx 7.2832$$

Answer b.

$$\text{Absolute maximum } f(2\pi) = 2\pi + 1 \approx 7.283 \text{ at } x = 2\pi$$

$$\text{Absolute minimum } f(-\pi) = -\pi - 1 \approx -4.142 \text{ at } x = -\pi$$

The critical number $x = \frac{\pi}{2}$ gives neither: it is a horizontal tangent on a curve that keeps rising.

Möbius: max $2\pi+1$ at $x=2\pi$; min $-\pi-1$ at $x=-\pi$

Q15.2 - Rectangle of least perimeter

Question. Find the dimensions of the rectangle of area 220 cm^2 with the smallest perimeter, and give that perimeter.

Step 1: name the variables and write both equations.

$$\text{Constraint: } xy = 220 \quad \text{Minimize: } P = 2x + 2y$$

Step 2: use the constraint to eliminate one variable.

$$y = \frac{220}{x} \implies P(x) = 2x + \frac{440}{x}, \quad x > 0$$

Step 3: differentiate.

$$P'(x) = 2 - \frac{440}{x^2}$$

Step 4: set to zero and solve.

$$2 = \frac{440}{x^2} \implies x^2 = 220 \implies x = \sqrt{220} = 2\sqrt{55}$$

Only the positive root is physical.

Step 5: confirm it is a minimum.

$$P''(x) = \frac{880}{x^3} > 0 \text{ for } x > 0$$

Concave up everywhere on the domain, so this critical point is the absolute minimum.

Step 6: find the other dimension.

$$y = \frac{220}{\sqrt{220}} = \sqrt{220} = 2\sqrt{55}$$

The rectangle is a **square**, which is the general result for fixed area and least perimeter.

Step 7: compute the perimeter.

$$P = 4\sqrt{220} = 8\sqrt{55}$$

Answer.

$$\text{Dimensions } 2\sqrt{55} \times 2\sqrt{55} \approx 14.83 \text{ cm} \times 14.83 \text{ cm}$$

$$\text{Minimum perimeter } 8\sqrt{55} \approx 59.33 \text{ cm}$$

Möbius: side $2\sqrt{55}$, perimeter $8\sqrt{55}$

Trap. Answering with the side length when the question also explicitly asks "What is the perimeter?". Both are marked.

Q15.3 - Absolute extremes by the Extreme Value Theorem

Question. Use the Extreme Value Theorem to find the absolute extreme values of

$$f(x) = \frac{1}{x^2 + 1} \quad \text{on } [-1, 1]$$

Step 1: check the theorem applies. The Extreme Value Theorem needs a continuous function on a closed, bounded interval. Here $x^2 + 1 \geq 1 > 0$, so the denominator never vanishes and f is continuous on all of $\mathbb{R}$, in particular on $[-1, 1]$. Therefore f **attains** both an absolute maximum and an absolute minimum on $[-1, 1]$.

Step 2: differentiate. Write $f(x) = (x^2 + 1)^{-1}$ and use the chain rule.

$$f'(x) = -(x^2 + 1)^{-2} (2x) = \frac{-2x}{(x^2 + 1)^2}$$

Step 3: find the critical numbers. The denominator is never zero, so f' is defined everywhere.

$$-2x = 0 \implies x = 0$$

Step 4: evaluate at the critical number and both endpoints.

$$f(-1) = \frac{1}{2}, \quad f(0) = \frac{1}{1} = 1, \quad f(1) = \frac{1}{2}$$

Step 5: compare.

Answer.

Absolute maximum $f(0) = 1$ at $x = 0$

Absolute minimum $f(-1) = f(1) = \frac{1}{2}$ at both $x = -1$ and $x = 1$

Möbius: max 1 at $x=0$; min $1/2$ at $x=-1$ and $x=1$

Note. The minimum is attained at two points. Reporting only one endpoint is an incomplete answer.

Q15.4 - Cheapest rectangular box

Question. A rectangular box has base length twice its width and volume 120 cm^3 . Material costs \$1.20 per cm^2 ; the lid costs \$1.50 per cm^2 . Find the dimensions of the cheapest box and the minimum cost.

Step 1: name the variables using the constraint immediately.

$$\text{width} = w, \quad \text{length} = 2w, \quad \text{height} = h$$

Step 2: write the volume constraint.

$$V = (2w)(w)(h) = 2w^2h = 120 \implies h = \frac{60}{w^2}$$

Step 3: itemise the surfaces, keeping the lid separate. This separation is the entire difficulty of the problem.

- base: $2w \cdot w = 2w^2$ square cm, charged at \$1.20
- four sides: $2(wh) + 2(2wh) = 6wh$ square cm, charged at \$1.20
- lid: $2w^2$ square cm, charged at \$1.50

Step 4: write the cost function.

$$C = 1.20(2w^2 + 6wh) + 1.50(2w^2) = 2.4w^2 + 7.2wh + 3w^2 = 5.4w^2 + 7.2wh$$

Step 5: eliminate h .

$$C(w) = 5.4w^2 + 7.2w \cdot \frac{60}{w^2} = 5.4w^2 + \frac{432}{w}, \quad w > 0$$

Step 6: differentiate and solve.

$$C'(w) = 10.8w - \frac{432}{w^2} = 0 \implies 10.8w^3 = 432 \implies w^3 = 40$$

$$w = \sqrt[3]{40} = 2\sqrt[3]{5} \approx 3.4200$$

Step 7: confirm it is a minimum.

$$C''(w) = 10.8 + \frac{864}{w^3} > 0 \text{ for } w > 0$$

Concave up on the whole domain, so this is the absolute minimum. Numerically $C''(\sqrt[3]{40}) = 32.4 > 0$.

Step 8: recover the other dimensions.

$$\text{length} = 2w = 2\sqrt[3]{40} \approx 6.8399$$

$$h = \frac{60}{w^2} = \frac{60}{40^{2/3}} = 3\sqrt[3]{5} \approx 5.1299$$

Step 9: compute the minimum cost.

$$C = 5.4(40^{2/3}) + \frac{432}{40^{1/3}} = \frac{324\sqrt[3]{25}}{5}$$

$$\approx 63.159 + 126.317 = 189.476$$

Answer.

$$\text{width } \sqrt[3]{40} \approx 3.42 \text{ cm, length } 2\sqrt[3]{40} \approx 6.84 \text{ cm, height } 3\sqrt[3]{5} \approx 5.13 \text{ cm}$$

Minimum cost: approximately \$189.48

Möbius: width $40^{1/3}$, length $2 \cdot 40^{1/3}$, height $3 \cdot 5^{1/3}$, cost $324 \cdot 5^{2/3} / 5$ or 189.48

Trap. Charging the lid at \$1.20 as well, or forgetting the base entirely. The box has a base at \$1.20, four sides at \$1.20, and a lid at \$1.50: six faces, two prices.

Q15.5 - Absolute extremes of a function with a cusp

Question. Find the absolute extreme values of

$$f(x) = (x^2 + 2x)^{2/3} \text{ on } [-2, 3]$$

Step 1: differentiate with the chain rule.

$$f'(x) = \frac{2}{3} (x^2 + 2x)^{-1/3} (2x + 2) = \frac{2(2x + 2)}{3\sqrt[3]{x^2 + 2x}}$$

Step 2: critical numbers where $f' = 0$.

$$2x + 2 = 0 \implies x = -1$$

Step 3: critical numbers where f' is undefined. The denominator vanishes when the inside is zero.

$$x^2 + 2x = 0 \implies x(x + 2) = 0 \implies x = 0, x = -2$$

At these points f **is** defined (it equals 0) but f' is not, so they are **cusps** and count as critical numbers. This is the feature the question is built around.

Step 4: evaluate at every critical number and both endpoints. Note the two-thirds power is (cube root)², so the result is never negative.

$$f(-2) = (4 - 4)^{2/3} = 0$$

$$f(-1) = (1 - 2)^{2/3} = (-1)^{2/3} = 1$$

$$f(0) = 0$$

$$f(3) = (9 + 6)^{2/3} = 15^{2/3} \approx 6.0822$$

Step 5: compare.

Answer.

Absolute maximum $f(3) = 15^{2/3} \approx 6.082$ at $x = 3$

Absolute minimum 0, attained at both $x = -2$ and $x = 0$

There is also a **local maximum** of 1 at $x = -1$, sitting between the two cusps.

Möbius: max $15^{2/3}$ at $x=3$; min 0 at $x=-2$ and $x=0$

Trap. Missing $x = 0$ and $x = -2$ because f' is undefined there rather than zero. A critical number is any point where f' is zero **or** undefined while f is defined, and here those points supply the minimum.

16. Antiderivatives and integration techniques

Q16.1 - Five integrals

Question. Compute each integral.

- a. $\int (x^2 - x) \sqrt{3x} \, dx$
- b. $\int \sin(2x) \cos x \, dx$
- c. $\int_a^b (x + \cos(2x)) \, dx$
- d. $\int (x^2 - 4)^2 \, dx$
- e. $\int_2^4 x \sqrt{x-1} \, dx$

Part a, step 1: pull the constant out of the root. There is no substitution here; the work is algebraic.

$$\sqrt{3x} = \sqrt{3} x^{1/2}$$

Part a, step 2: distribute.

$$(x^2 - x) \sqrt{3} x^{1/2} = \sqrt{3} (x^{5/2} - x^{3/2})$$

Part a, step 3: apply the power rule to each term.

$$\int x^{5/2} dx = \frac{2}{7} x^{7/2}, \quad \int x^{3/2} dx = \frac{2}{5} x^{5/2}$$

Answer a.

$$\int (x^2 - x) \sqrt{3} x dx = \sqrt{3} \left(\frac{2}{7} x^{7/2} - \frac{2}{5} x^{5/2} \right) + C$$

Möbius: $\text{sqrt}(3) * ((2/7) * x^{(7/2)} - (2/5) * x^{(5/2)}) + C$

Part b, step 1: remove the double angle. The two different arguments, $2x$ and x , block substitution, so use $\sin(2x) = 2 \sin x \cos x$.

$$\int \sin(2x) \cos x dx = \int 2 \sin x \cos^2 x dx$$

Part b, step 2: substitute.

$$u = \cos x, \quad du = -\sin x dx$$

$$= 2 \int \cos^2 x \sin x dx = -2 \int u^2 du$$

Part b, step 3: integrate and back-substitute.

$$= -\frac{2u^3}{3} + C = -\frac{2}{3} \cos^3 x + C$$

Answer b.

$$\int \sin(2x) \cos x dx = -\frac{2}{3} \cos^3 x + C$$

Möbius: $-(2/3) * \cos(x)^3 + C$

Check. $\frac{d}{dx} \left(-\frac{2}{3} \cos^3 x \right) = -2 \cos^2 x (-\sin x) = 2 \sin x \cos^2 x = \sin(2x) \cos x$. Confirmed.

Part c, step 1: integrate term by term with symbolic limits.

$$\int x dx = \frac{x^2}{2}, \quad \int \cos(2x) dx = \frac{\sin(2x)}{2}$$

The $\frac{1}{2}$ on the sine comes from the inside derivative of $2x$.

Part c, step 2: apply the limits.

$$\left[\frac{x^2}{2} + \frac{\sin(2x)}{2} \right]_a^b$$

Answer c.

$$\int_a^b (x + \cos(2x)) dx = \frac{b^2 - a^2}{2} + \frac{\sin(2b) - \sin(2a)}{2}$$

Möbius: $(b^2 - a^2)/2 + (\sin(2b) - \sin(2a))/2$

Trap. This is a **definite** integral, so no $+C$. The limits are letters, but it is still a number once a and b are known.

Part d, step 1: expand first. Do not reach for substitution; there is no inner derivative available.

$$(x^2 - 4)^2 = x^4 - 8x^2 + 16$$

Part d, step 2: integrate term by term.

Answer d.

$$\int (x^2 - 4)^2 dx = \frac{x^5}{5} - \frac{8x^3}{3} + 16x + C$$

Möbius: $x^5/5 - 8x^3/3 + 16x + C$

Trap. Writing $\frac{(x^2 - 4)^3}{3}$ by treating the bracket as if the chain rule ran backwards. That requires the inner derivative $2x$ to be present, and it is not.

Part e, step 1: substitute.

$$u = x - 1 \implies x = u + 1, \quad dx = du$$

Part e, step 2: change the limits. For a definite integral, convert the limits and never convert back.

$$x = 2 \implies u = 1, \quad x = 4 \implies u = 3$$

Part e, step 3: rewrite the integrand.

$$\int_1^3 (u + 1)u^{1/2} du = \int_1^3 (u^{3/2} + u^{1/2}) du$$

Part e, step 4: integrate.

$$= \left[\frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right]_1^3$$

Part e, step 5: evaluate, using $3^{5/2} = 9\sqrt{3}$ and $3^{3/2} = 3\sqrt{3}$.

$$\text{at } u = 3 : \frac{2}{5}(9\sqrt{3}) + \frac{2}{3}(3\sqrt{3}) = \frac{18\sqrt{3}}{5} + 2\sqrt{3} = \frac{28\sqrt{3}}{5}$$

$$\text{at } u = 1 : \frac{2}{5} + \frac{2}{3} = \frac{16}{15}$$

Part e, step 6: subtract.

$$\frac{28\sqrt{3}}{5} - \frac{16}{15} = \frac{84\sqrt{3} - 16}{15}$$

Answer e.

$$\int_2^4 x\sqrt{x-1} dx = \frac{84\sqrt{3} - 16}{15} \approx 8.6328$$

Möbius: $(84*\text{sqrt}(3) - 16)/15$

Trap. Substituting $u = x - 1$ but leaving the stray x in the integrand. You must also replace x by $u + 1$.

Q16.2 - Four integrals, naming the technique

Question. Integrate each, indicating the technique used.

- a. $\int \frac{\cos(\sqrt{2x})}{\sqrt{x}} dx$
- b. $\int_0^{\pi/3} \tan x \sec^2 x dx$
- c. $\int \frac{x^3 + \sqrt{5x} - 4}{x^2} dx$
- d. $\int \sec^3 x \tan x dx$, with the hint $\sec^3 x \tan x = \sec^2 x \sec x \tan x$

Part a. Technique: u-substitution.

Step 1: choose u as the inside of the cosine.

$$u = \sqrt{2x} = \sqrt{2} x^{1/2}$$

Step 2: differentiate.

$$du = \frac{\sqrt{2}}{2} x^{-1/2} dx = \frac{\sqrt{2}}{2\sqrt{x}} dx$$

Step 3: solve for the piece that appears in the integral.

$$\frac{dx}{\sqrt{x}} = \frac{2}{\sqrt{2}} du = \sqrt{2} du$$

Step 4: rewrite and integrate.

$$\int \cos(u) \sqrt{2} du = \sqrt{2} \sin(u) + C$$

Answer a.

$$\int \frac{\cos(\sqrt{2x})}{\sqrt{x}} dx = \sqrt{2} \sin(\sqrt{2x}) + C$$

Möbius: $\text{sqrt}(2)*\sin(\text{sqrt}(2*x))+C$

Check. $\frac{d}{dx} \sqrt{2} \sin(\sqrt{2x}) = \sqrt{2} \cos(\sqrt{2x}) \cdot \frac{\sqrt{2}}{2\sqrt{x}} = \frac{\cos(\sqrt{2x})}{\sqrt{x}}$. Confirmed.

Part b. Technique: u-substitution with changed limits.

Step 1: choose u . $\sec^2 x$ is exactly the derivative of $\tan x$, which is the signal.

$$u = \tan x, \quad du = \sec^2 x \, dx$$

Step 2: change the limits.

$$x = 0 \implies u = \tan 0 = 0$$

$$x = \frac{\pi}{3} \implies u = \tan \frac{\pi}{3} = \sqrt{3}$$

Step 3: integrate.

$$\int_0^{\sqrt{3}} u \, du = \left[\frac{u^2}{2} \right]_0^{\sqrt{3}} = \frac{3}{2} - 0$$

Answer b.

$$\int_0^{\pi/3} \tan x \sec^2 x \, dx = \frac{3}{2}$$

Möbius: 3/2

Part c. Technique: algebraic rewriting, then the power rule.

Step 1: split the fraction term by term.

$$\frac{x^3}{x^2} = x, \quad \frac{\sqrt{5}x}{x^2} = \frac{\sqrt{5}x^{1/2}}{x^2} = \sqrt{5}x^{-3/2}, \quad \frac{-4}{x^2} = -4x^{-2}$$

Step 2: integrate each piece.

$$\int x \, dx = \frac{x^2}{2}$$

$$\int \sqrt{5}x^{-3/2} \, dx = \sqrt{5} \cdot \frac{x^{-1/2}}{-\frac{1}{2}} = -2\sqrt{5}x^{-1/2}$$

$$\int -4x^{-2} \, dx = -4 \cdot \frac{x^{-1}}{-1} = \frac{4}{x}$$

Answer c.

$$\int \frac{x^3 + \sqrt{5}x - 4}{x^2} \, dx = \frac{x^2}{2} - \frac{2\sqrt{5}}{\sqrt{x}} + \frac{4}{x} + C$$

Möbius: $x^2/2 - 2\sqrt{5}/\sqrt{x} + 4/x + C$

Trap. Trying to integrate the numerator and denominator separately. There is no quotient rule for integrals; divide through first.

Part d. Technique: u-substitution, guided by the hint.

Step 1: read the hint as a split.

$$\sec^3 x \tan x = \sec^2 x \cdot (\sec x \tan x)$$

Step 2: choose u so the bracketed part is du .

$$u = \sec x, \quad du = \sec x \tan x \, dx$$

Step 3: rewrite.

$$\int \sec^2 x (\sec x \tan x \, dx) = \int u^2 du$$

Step 4: integrate and back-substitute.

$$= \frac{u^3}{3} + C = \frac{\sec^3 x}{3} + C$$

Answer d.

$$\int \sec^3 x \tan x \, dx = \frac{\sec^3 x}{3} + C$$

Möbius: $\sec(x)^3/3 + C$

Check. $\frac{d}{dx} \frac{\sec^3 x}{3} = \sec^2 x \cdot \sec x \tan x = \sec^3 x \tan x$. Confirmed.

17. Properties of the definite integral

All three of these use the same bounding property, so learn it once:

$$\text{if } m \leq f(x) \leq M \text{ on } [a, b], \text{ then } m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a)$$

Q17.1 - Interval containing a secant integral

Question. Use the properties of the integral to find an interval containing

$$\int_0^{\pi/3} \sec x \, dx$$

Step 1: identify the interval and its width.

$$[a, b] = \left[0, \frac{\pi}{3}\right], \quad b - a = \frac{\pi}{3}$$

Step 2: find the extreme values of $\sec x$ on that interval. On $\left[0, \frac{\pi}{3}\right]$, cosine decreases from 1 to $\frac{1}{2}$, so its reciprocal $\sec x$ **increases** from 1 to 2.

$$\sec 0 = 1 \quad (\text{minimum}), \quad \sec \frac{\pi}{3} = 2 \quad (\text{maximum})$$

The interval stops short of $\frac{\pi}{2}$, so $\sec x$ stays finite and this is legitimate.

Step 3: apply the bounding property.

$$1 \cdot \frac{\pi}{3} \leq \int_0^{\pi/3} \sec x \, dx \leq 2 \cdot \frac{\pi}{3}$$

Answer.

$$\frac{\pi}{3} \leq \int_0^{\pi/3} \sec x \, dx \leq \frac{2\pi}{3}$$

Numerically $1.047 \leq I \leq 2.094$.

Möbius: interval $[\pi/3, 2\pi/3]$

Verification. The exact value is $\ln(2 + \sqrt{3}) \approx 1.3170$, which does sit inside $[1.047, 2.094]$.

Q17.2 - Estimate an integral from its properties

Question. Use the properties of the definite integral to estimate

$$\int_{-1}^1 \frac{1}{x^2 + 1} \, dx$$

Step 1: identify the interval and its width.

$$[a, b] = [-1, 1], \quad b - a = 2$$

The width is 2, not 1; this is the most common slip on this question.

Step 2: find the extreme values of the integrand. This is the same function as Q15.3, so reuse that analysis. $f(x) = \frac{1}{x^2 + 1}$ is largest where x^2 is smallest.

$$\text{maximum } f(0) = 1, \quad \text{minimum } f(\pm 1) = \frac{1}{2}$$

Step 3: apply the bounding property.

$$\frac{1}{2}(2) \leq \int_{-1}^1 \frac{dx}{x^2 + 1} \leq 1(2)$$

Answer.

$$1 \leq \int_{-1}^1 \frac{1}{x^2 + 1} \, dx \leq 2$$

Möbius: interval $[1, 2]$

Verification. The exact value is $\arctan(1) - \arctan(-1) = \frac{\pi}{2} \approx 1.5708$, comfortably inside $[1, 2]$.

Q17.3 - Bound an integral from its properties

Question. Use the properties of definite integrals to bound

$$\int_{-2}^3 (x^2 + 2x)^{2/3} dx$$

Step 1: identify the interval and its width.

$$[a, b] = [-2, 3], \quad b - a = 5$$

Step 2: find the extreme values of the integrand. This is exactly the function from Q15.5 on exactly that interval, so the work is already done.

$$\text{minimum } 0 \text{ at } x = -2 \text{ and } x = 0, \quad \text{maximum } 15^{2/3} \text{ at } x = 3$$

Step 3: apply the bounding property.

$$0 \cdot 5 \leq \int_{-2}^3 (x^2 + 2x)^{2/3} dx \leq 15^{2/3} \cdot 5$$

Answer.

$$0 \leq \int_{-2}^3 (x^2 + 2x)^{2/3} dx \leq 5 \cdot 15^{2/3} \approx 30.41$$

Möbius: interval $[0, 5 \cdot 15^{2/3}]$

Verification. The true value is about 10.67, which lies inside $[0, 30.41]$. The bound is loose because the integrand only reaches its maximum at the single right endpoint, which is normal for this method: it gives a guaranteed range, not a sharp estimate.

Trap. Concluding the lower bound is negative. The exponent $\frac{2}{3}$ is an even power of a cube root, so the integrand is never negative, and 0 is the correct floor.

18. The Fundamental Theorem of Calculus

Q18.1 - Differentiate an integral with variable limits

Question. Use the Fundamental Theorem of Calculus to evaluate

$$\frac{d}{dx} \int_{2x}^x \sin(t^2) dt$$

Step 1: recall the basic form of FTC part 1.

$$\frac{d}{dx} \int_a^x f(t) dt = f(x), \quad a \text{ constant}$$

Here **both** limits are variable, so the basic form does not apply directly.

Step 2: split at a constant. Introduce any convenient constant c and use the additivity of the integral.

$$\int_{2x}^x = \int_{2x}^c + \int_c^x = \int_c^x - \int_c^{2x}$$

Step 3: differentiate the first piece. The upper limit is exactly x , so this is the basic form.

$$\frac{d}{dx} \int_c^x \sin(t^2) dt = \sin(x^2)$$

Step 4: differentiate the second piece with the chain rule. The upper limit is $2x$, so evaluate the integrand at $2x$ and multiply by the derivative of $2x$.

$$\frac{d}{dx} \int_c^{2x} \sin(t^2) dt = \sin((2x)^2) \cdot 2 = 2 \sin(4x^2)$$

Step 5: subtract.

Answer.

$$\frac{d}{dx} \int_{2x}^x \sin(t^2) dt = \sin(x^2) - 2 \sin(4x^2)$$

Möbius: $\sin(x^2) - 2 \sin(4x^2)$

General rule worth memorizing.

$$\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h(x))h'(x) - f(g(x))g'(x)$$

Trap. Forgetting that $(2x)^2 = 4x^2$, not $2x^2$, and forgetting the factor 2 from the chain rule. Both errors hide in the same term.

19. Applications of integration

Q19.1 - Area between a line and a parabola

Question. Find the area between $y = x$ and $y = 2 - x^2$.

Step 1: find the intersection points. No interval is given, so the curves themselves supply the limits.

$$x = 2 - x^2 \implies x^2 + x - 2 = 0 \implies (x + 2)(x - 1) = 0$$

$$x = -2, \quad x = 1$$

Step 2: decide which curve is on top. Test any point strictly between, say $x = 0$.

$$\text{parabola: } 2 - 0 = 2, \quad \text{line: } 0$$

The parabola is above the line on all of $(-2, 1)$, and the curves meet only at the endpoints, so no split is needed.

Step 3: set up the integral of top minus bottom.

$$A = \int_{-2}^1 [(2 - x^2) - x] dx = \int_{-2}^1 (2 - x - x^2) dx$$

Step 4: antidifferentiate.

$$= \left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1$$

Step 5: evaluate at the top limit.

$$2 - \frac{1}{2} - \frac{1}{3} = \frac{12 - 3 - 2}{6} = \frac{7}{6}$$

Step 6: evaluate at the bottom limit.

$$-4 - \frac{4}{2} - \frac{-8}{3} = -4 - 2 + \frac{8}{3} = -6 + \frac{8}{3} = -\frac{10}{3}$$

Step 7: subtract.

$$A = \frac{7}{6} - \left(-\frac{10}{3} \right) = \frac{7}{6} + \frac{20}{6} = \frac{27}{6} = \frac{9}{2}$$

Answer.

$$A = \frac{9}{2} = 4.5 \text{ square units}$$

Möbius: 9/2

Trap. Integrating line minus parabola, which returns $-\frac{9}{2}$. An area is positive; if you get a negative number you had the order backwards.

Q19.2 - Area between a cosine and a horizontal line

Question. Find the area between $y = \cos x$ and the horizontal line $y = \frac{1}{\sqrt{2}}$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Step 1: find where the curves cross.

$$\cos x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \implies x = \pm \frac{\pi}{4}$$

Both crossings lie strictly inside the given interval, so the integral **must** be split.

Step 2: decide which is on top in each piece. Test one point in each region.

$$x = 0 : \cos 0 = 1 > \frac{1}{\sqrt{2}} \implies \text{cosine on top on } \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$$

$$x = \frac{\pi}{2} : \cos \frac{\pi}{2} = 0 < \frac{1}{\sqrt{2}} \implies \text{line on top on } \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

By symmetry the same holds on $\left(-\frac{\pi}{2}, -\frac{\pi}{4}\right)$.

Step 3: write the total area as three integrals.

$$A = \int_{-\pi/2}^{-\pi/4} \left(\frac{1}{\sqrt{2}} - \cos x \right) dx + \int_{-\pi/4}^{\pi/4} \left(\cos x - \frac{1}{\sqrt{2}} \right) dx + \int_{\pi/4}^{\pi/2} \left(\frac{1}{\sqrt{2}} - \cos x \right) dx$$

Step 4: evaluate the middle piece. The antiderivative is $\sin x - \frac{x}{\sqrt{2}}$.

$$\left[\sin x - \frac{x}{\sqrt{2}} \right]_{-\pi/4}^{\pi/4} = \left(\frac{\sqrt{2}}{2} - \frac{\pi}{4\sqrt{2}} \right) - \left(-\frac{\sqrt{2}}{2} + \frac{\pi}{4\sqrt{2}} \right) = \sqrt{2} - \frac{\pi}{2\sqrt{2}}$$

Numerically $1.41421 - 1.11072 = 0.30349$.

Step 5: evaluate the right piece. The antiderivative is $\frac{x}{\sqrt{2}} - \sin x$.

$$\left[\frac{x}{\sqrt{2}} - \sin x \right]_{\pi/4}^{\pi/2} = \left(\frac{\pi}{2\sqrt{2}} - 1 \right) - \left(\frac{\pi}{4\sqrt{2}} - \frac{\sqrt{2}}{2} \right) = \frac{\pi}{4\sqrt{2}} - 1 + \frac{\sqrt{2}}{2}$$

Numerically $0.55536 - 1 + 0.70711 = 0.26247$.

Step 6: the left piece equals the right by symmetry. Both $\cos x$ and the constant line are even, so the region is symmetric about the y -axis.

$$\text{left piece} = 0.26247$$

Step 7: add all three. The π terms cancel exactly between the middle and outer pieces.

$$A = \left(\sqrt{2} - \frac{\pi}{2\sqrt{2}} \right) + 2 \left(\frac{\pi}{4\sqrt{2}} - 1 + \frac{\sqrt{2}}{2} \right) = \sqrt{2} + \sqrt{2} - 2$$

Answer.

$$A = 2\sqrt{2} - 2 \approx 0.8284 \text{ square units}$$

Möbius: $2*\text{sqrt}(2) - 2$

Note on reading the question. Because an explicit interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is given, the answer is the **total** area over that whole interval, which is $2\sqrt{2} - 2$. If a version of this question instead asks only for the region **enclosed** by the two curves, that is the middle piece alone:

$$\sqrt{2} - \frac{\pi}{2\sqrt{2}} = \sqrt{2} - \frac{\pi\sqrt{2}}{4} \approx 0.3035$$

Read which one is wanted before you set up.

Trap. Integrating $\cos x - \frac{1}{\sqrt{2}}$ straight across $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ without splitting. That returns $2 - \frac{\pi}{\sqrt{2}} \approx -0.221$, a signed value in which the outer negative regions cancel part of the middle positive region.

Q19.3 - Net change, water flowing from a tank

Question. Water flows from a tank at $r(t) = 180 - 6t$ litres per minute for $0 \leq t \leq 50$. Find the amount that flows out during the first 15 minutes.

Step 1: recognise the question type. A **rate** is given and a **total** is wanted, so integrate.

$$\text{Amount} = \int_0^{15} r(t) dt$$

Step 2: set up.

$$= \int_0^{15} (180 - 6t) dt$$

Step 3: antidifferentiate.

$$= [180t - 3t^2]_0^{15}$$

Step 4: evaluate.

$$= (180(15) - 3(225)) - 0 = 2700 - 675 = 2025$$

Answer.

2025 litres

Möbius: 2025

Sanity check. The rate starts at 180 L/min and falls to $180 - 90 = 90$ L/min at $t = 15$. Since the rate is linear, the average is $\frac{180 + 90}{2} = 135$ L/min, and $135 \times 15 = 2025$. The two methods agree.

Q19.4 - Work, raising one end of a chain

Question. A chain lying on the ground is 10 m long with mass 80 kg. Find the work required to raise one end to a height of 6 m.

Step 1: find the linear density.

$$\rho = \frac{80 \text{ kg}}{10 \text{ m}} = 8 \text{ kg/m}$$

Step 2: picture the final configuration. Raising one end to 6 m lifts only the top 6 m of chain into the air; the remaining 4 m stays flat on the ground and is never lifted. This is why the integral runs to 6, not to 10.

Step 3: slice by final height. Let y be the final height of a small piece, $0 \leq y \leq 6$. The piece of chain that ends at height y started on the ground, so it is lifted a distance y .

$$dm = \rho dy = 8 dy$$

$$dW = (dm)gy = 8(9.8)y dy = 78.4 y dy$$

Step 4: integrate.

$$W = \int_0^6 78.4 y \, dy = 78.4 \left[\frac{y^2}{2} \right]_0^6 = 78.4 \cdot 18$$

Answer.

$$W = 1411.2 \text{ J}$$

Möbius: 1411.2 (using $g = 9.8$)

Note. With $g = 9.81$ the answer is 1412.64 J. State which g you used.

Trap. Integrating from 0 to 10, which treats the whole chain as lifted and gives 3920 J. Only 6 m of chain leaves the ground.

Q19.5 - Work, stretching a spring

Question. 20 ft-lb of work stretches a spring 1 ft beyond its natural length.

- a. Find the spring constant.
- b. Find the work needed to stretch it 2 ft beyond natural length.

Part a, step 1: write Hooke's Law and the work integral.

$$F(x) = kx, \quad W = \int_0^d kx \, dx = \frac{1}{2}kd^2$$

Part a, step 2: substitute the given data, $W = 20$ and $d = 1$.

$$20 = \frac{1}{2}k(1)^2 = \frac{k}{2}$$

Part a, step 3: solve.

$$k = 40$$

Answer a.

$$k = 40 \text{ lb/ft}$$

Möbius: 40

Part b, step 1: use the same formula with $d = 2$.

$$W = \int_0^2 40x \, dx = \left[20x^2 \right]_0^2 = 20(4)$$

Answer b.

$$W = 80 \text{ ft-lb}$$

Möbius: 80

Note. Doubling the stretch **quadruples** the work, because $W \propto d^2$. Answering 40 ft-lb by assuming proportionality is the trap this question is built to catch.

Q19.6 - Average temperature of a metal rod

Question. A rod 6 m long has temperature $3x$ degrees centigrade at distance x metres from one end. Find the average temperature.

Step 1: write the average value formula.

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Step 2: identify the pieces.

$$f(x) = 3x, \quad [a, b] = [0, 6], \quad b - a = 6$$

Step 3: integrate.

$$\int_0^6 3x dx = \left[\frac{3x^2}{2} \right]_0^6 = \frac{3(36)}{2} = 54$$

Step 4: divide by the width.

$$f_{\text{avg}} = \frac{54}{6} = 9$$

Answer.

$$9^\circ \text{C}$$

Möbius: 9

Sanity check. The temperature runs linearly from 0 at one end to 18 at the other, and the average of a linear function is its midpoint value: $\frac{0+18}{2} = 9$. Confirmed.

Q19.7 - Find k from an average value

Question. Find a positive number k such that the average value of $f(x) = \frac{5}{x^2}$ over the interval between 1 and k is 32.

Step 1: write the average value equation.

$$\frac{1}{k-1} \int_1^k \frac{5}{x^2} dx = 32$$

Step 2: integrate.

$$\int_1^k 5x^{-2} dx = \left[-\frac{5}{x} \right]_1^k = -\frac{5}{k} + 5 = 5 \left(1 - \frac{1}{k} \right)$$

Step 3: simplify the numerator into a single fraction. This is the step that makes the whole problem collapse.

$$5 \left(\frac{k-1}{k} \right)$$

Step 4: divide by the width and watch $(k-1)$ cancel.

$$\frac{1}{k-1} \cdot \frac{5(k-1)}{k} = \frac{5}{k}$$

Step 5: solve.

$$\frac{5}{k} = 32 \implies k = \frac{5}{32}$$

Answer.

$$k = \frac{5}{32} = 0.15625$$

Möbius: $5/32$

Note. k is positive, as required, but it is **less than 1**, so the interval is really $\left[\frac{5}{32}, 1 \right]$. The phrase "between 1 and k " allows this, and the algebra is identical either way: reversing the limits flips the sign of both the integral and the width, and the two flips cancel.

Verification. On $\left[\frac{5}{32}, 1 \right]$, the average of $\frac{5}{x^2}$ is exactly 32, confirmed by direct computation.

Trap. Rejecting $k = \frac{5}{32}$ because it is smaller than 1 and hunting for a larger root. There is no other solution: the average simplifies to exactly $\frac{5}{k}$, which is one-to-one.

Q19.8 - Displacement and distance travelled

Question. A particle moves with velocity $v(t) = t^2 - 3t + 2$ m/s.

- a. Find the displacement on $[0, 3]$.
- b. Find the distance travelled on $[0, 3]$.

Part a, step 1: displacement is the plain integral of velocity. Signs are kept, so backward motion subtracts.

$$s = \int_0^3 (t^2 - 3t + 2) dt$$

Part a, step 2: antidifferentiate and evaluate.

$$= \left[\frac{t^3}{3} - \frac{3t^2}{2} + 2t \right]_0^3 = 9 - \frac{27}{2} + 6 = 15 - 13.5 = \frac{3}{2}$$

Answer a.

$$\text{Displacement} = \frac{3}{2} = 1.5 \text{ m}$$

Möbius: 3/2**Part b, step 1: distance needs the absolute value.**

$$d = \int_0^3 |v(t)| dt$$

Part b, step 2: find where the velocity changes sign.

$$t^2 - 3t + 2 = (t - 1)(t - 2) = 0 \implies t = 1, t = 2$$

Both lie inside $[0, 3]$, so split there.**Part b, step 3: determine the sign on each piece.**

$$t = 0.5 : v > 0 \quad t = 1.5 : v < 0 \quad t = 2.5 : v > 0$$

The particle moves forward, reverses, then moves forward again.

Part b, step 4: integrate each piece.

$$\int_0^1 v dt = \frac{1}{3} - \frac{3}{2} + 2 = \frac{5}{6}$$

$$\int_1^2 v dt = -\frac{1}{6}$$

$$\int_2^3 v dt = \frac{5}{6}$$

Part b, step 5: add the absolute values.

$$d = \frac{5}{6} + \left| -\frac{1}{6} \right| + \frac{5}{6} = \frac{5 + 1 + 5}{6} = \frac{11}{6}$$

Answer b.

$$\text{Distance} = \frac{11}{6} \approx 1.833 \text{ m}$$

Möbius: 11/6

Check. The three signed pieces sum to $\frac{5}{6} - \frac{1}{6} + \frac{5}{6} = \frac{9}{6} = \frac{3}{2}$, matching part a. Distance always exceeds displacement when the velocity changes sign, and $\frac{11}{6} > \frac{3}{2}$ as expected.

Q19.9 - Sprinter with piecewise acceleration

Question. A sprinter in a 100 m race accelerates at 4 m/s^2 for 2 seconds, then accelerates at zero for the rest of the race. Find her time for the race.

Step 1: phase 1 velocity. Integrate the acceleration, with $v(0) = 0$ from a standing start.

$$v(t) = \int 4 \, dt = 4t \implies v(2) = 8 \text{ m/s}$$

Step 2: phase 1 distance. Integrate the velocity over the first 2 seconds.

$$d_1 = \int_0^2 4t \, dt = \left[2t^2 \right]_0^2 = 8 \text{ m}$$

Step 3: phase 2 setup. Zero acceleration means constant velocity, and it holds at the value reached at the end of phase 1.

$$v = 8 \text{ m/s for the rest of the race}$$

Step 4: remaining distance.

$$100 - 8 = 92 \text{ m}$$

Step 5: time for phase 2.

$$t_2 = \frac{92}{8} = 11.5 \text{ s}$$

Step 6: total time.

$$T = 2 + 11.5 = 13.5 \text{ s}$$

Answer.

$$T = 13.5 \text{ seconds}$$

Möbius: 13.5

Trap. Dividing the full 100 m by 8 m/s to get 12.5 s. That ignores the 8 m already covered while accelerating, and it ignores that she was slower than 8 m/s for those first 2 seconds.

Part 6: The one-page formula card

Memorize these before each exam. Each entry gives the Möbius typing form first and the printed mathematical form under it.

Derivative rules:

- Constant: $d/dx [c] = 0$.
- Math: $\frac{d}{dx}(c) = 0$
- Power: $d/dx [x^n] = n x^{(n - 1)}$.

- Math: $\frac{d}{dx}(x^n) = nx^{n-1}$
- Sum: $(f + g)' = f' + g'$.
 - Math: $(f + g)' = f' + g'$
- Product: $(fg)' = f'g + fg'$.
 - Math: $(fg)' = f'g + fg'$
- Quotient: $(f/g)' = (f'g - fg') / g^2$.
 - Math: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- Chain: $d/dx [f(g(x))] = f'(g(x)) g'(x)$.
 - Math: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$

Trig derivatives:

- $\sin x \rightarrow \cos x$.
 - Math: $\frac{d}{dx} \sin x = \cos x$
- $\cos x \rightarrow -\sin x$.
 - Math: $\frac{d}{dx} \cos x = -\sin x$
- $\tan x \rightarrow \sec^2 x$.
 - Math: $\frac{d}{dx} \tan x = \sec^2 x$
- $\cot x \rightarrow -\csc^2 x$.
 - Math: $\frac{d}{dx} \cot x = -\csc^2 x$
- $\sec x \rightarrow \sec x \tan x$.
 - Math: $\frac{d}{dx} \sec x = \sec x \tan x$
- $\csc x \rightarrow -\csc x \cot x$.
 - Math: $\frac{d}{dx} \csc x = -\csc x \cot x$

Exponential and log:

- $e^x \rightarrow e^x$.
 - Math: $\frac{d}{dx} e^x = e^x$
- $a^x \rightarrow a^x \ln a$.
 - Math: $\frac{d}{dx} a^x = a^x \ln a$
- $\ln x \rightarrow 1/x$.
 - Math: $\frac{d}{dx} \ln x = \frac{1}{x}$

Integrals:

- $\int x^n dx = x^{(n+1)} / (n+1) + C$, for $n \neq -1$.
 - Math: $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
- $\int 1/x dx = \ln|x| + C$.
 - Math: $\int \frac{1}{x} dx = \ln|x| + C$
- $\int \sin x dx = -\cos x + C$.
 - Math: $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$.
 - Math: $\int \cos x dx = \sin x + C$
- $\int \sec^2 x dx = \tan x + C$.

- Math: $\int \sec^2 x \, dx = \tan x + C$
- $\int \csc^2 x \, dx = -\cot x + C.$
 - Math: $\int \csc^2 x \, dx = -\cot x + C$
- $\int \sec x \tan x \, dx = \sec x + C.$
 - Math: $\int \sec x \tan x \, dx = \sec x + C$
- $\int \csc x \cot x \, dx = -\csc x + C.$
 - Math: $\int \csc x \cot x \, dx = -\csc x + C$

Fundamental Theorem:

- $d/dx \int [a \text{ to } x] f(t) \, dt = f(x).$
 - Math: $\frac{d}{dx} \int_a^x f(t) \, dt = f(x)$
- $\int [a \text{ to } b] f(x) \, dx = F(b) - F(a).$
 - Math: $\int_a^b f(x) \, dx = F(b) - F(a)$
- Both limits variable: $d/dx \int [g(x) \text{ to } h(x)] f(t) \, dt = f(h(x)) h'(x) - f(g(x)) g'(x).$
 - Math: $\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) \, dt = f(h(x))h'(x) - f(g(x))g'(x)$

Average value:

- $f_{\text{avg}} = (1 / (b - a)) \int [a \text{ to } b] f(x) \, dx.$
 - Math: $f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx$

Bounding a definite integral:

- $m(b - a) \leq \int [a \text{ to } b] f \leq M(b - a).$
 - Math: $m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a)$ when $m \leq f \leq M$

Special limits:

- $\lim(u \rightarrow 0) \sin(u)/u = 1.$
 - Math: $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$
- $\lim(u \rightarrow 0) (1 - \cos(u))/u = 0.$
 - Math: $\lim_{u \rightarrow 0} \frac{1 - \cos u}{u} = 0$

Key identities:

- $\sin^2 x + \cos^2 x = 1.$
 - Math: $\sin^2 x + \cos^2 x = 1$
- $1 + \tan^2 x = \sec^2 x.$
 - Math: $1 + \tan^2 x = \sec^2 x$
- $1 + \cot^2 x = \csc^2 x.$
 - Math: $1 + \cot^2 x = \csc^2 x$
- $\sin(2x) = 2 \sin x \cos x.$
 - Math: $\sin(2x) = 2 \sin x \cos x$
- $\cos(2x) = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1.$
 - Math: $\cos(2x) = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1$
- $1 - \cos(2x) = 2 \sin^2 x.$
 - Math: $1 - \cos(2x) = 2 \sin^2 x$
- $1 + \cos(2x) = 2 \cos^2 x.$
 - Math: $1 + \cos(2x) = 2 \cos^2 x$

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$.
- Math: $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$.
- Math: $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Applications:

- Spring work: $W = (1/2) k d^2$.
- Math: $W = \frac{1}{2} k d^2$
 - Chain work: $W = \int \rho g y \, dy$.
- Math: $W = \int \rho g y \, dy$
 - Displacement: $\int v \, dt$, distance: $\int |v| \, dt$.
- Math: $\int_a^b v \, dt$ and $\int_a^b |v| \, dt$
 - Newton's method: $x_{n+1} = x_n - f(x_n)/f'(x_n)$.
- Math: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
 - Linearization: $f(a + dx) \approx f(a) + f'(a) dx$.
- Math: $f(a + dx) \approx f(a) + f'(a) dx$
 - Cone volume: $V = \pi r^2 h / 3$.
- Math: $V = \frac{\pi r^2 h}{3}$
-

Part 7: Practice map

Use this map to drill every question note against the topic. Every question listed here is fully worked in Part 5.

Foundations:

- Exact trig values: Q01.1.
- Composite domains: Q01.2.
- Function definition: Q01.3.
- Modeling a cone: Q01.4.
- Transformations: Q02.1, Q02.2.

Limits and graphs:

- Limits: Q03.1, Q03.2, Q03.3.
- Asymptotes: Q04.1.
- Sketching from conditions: Q05.1, Q05.2, Q05.3.
- Rate interpretation: Q06.1.

Derivatives:

- Definition of derivative: Q07.1.
- Derivative rules: Q08.1, Q08.2, Q08.3, Q08.4.
- Tangent line: Q09.1.
- Implicit differentiation: Q10.1, Q10.2.
- Related rates: Q11.1, Q11.2.
- Differentials: Q12.1, Q12.2, Q12.3.
- Newton's method: Q13.1.
- Curve sketching: Q14.1.
- Extrema and optimization: Q15.1 through Q15.5.

Integration:

- Antiderivatives: Q16.1, Q16.2.

- Bounding integrals: Q17.1, Q17.2, Q17.3.
- Fundamental Theorem: Q18.1.
- Applications: Q19.1 through Q19.9.

Questions that reuse each other, so learn them as pairs:

- Q15.3 and Q17.2 use the same function $\frac{1}{x^2 + 1}$ on the same interval.
- Q15.5 and Q17.3 use the same function $(x^2 + 2x)^{2/3}$ on the same interval.
- Q12.2 and Q12.3a are the same linearization at 60° , once for sine and once for cosine.
- Q14.1b and Q15.1b are $x + \sin x$ and $x + \cos x$, and both hinge on the derivative never going negative.

Sample exams:

- Sample Midterm Exam 1 and Sample Midterm Exam 2 in the course folder.
 - Sample Final Exam 1 and Sample Final Exam 2 in the course folder.
 - The all-exam question bank is MATH265-All-Exam-Questions.md.
-

Part 8: Weekly practice plan

Week 1:

- Master exact trig values and domains.
- Do Q01.1 through Q01.4.

Week 2:

- Master limits, asymptotes, and graph transformations.
- Do Q02.1 through Q05.3.

Week 3:

- Master derivative rules and the chain rule.
- Do Q06.1 through Q08.4.

Week 4:

- Master tangent lines, implicit differentiation, related rates, and differentials.
- Do Q09.1 through Q12.3.
- Take Sample Midterm 1 and 2 under timed conditions.

Week 5:

- Master curve sketching, extrema, and optimization.
- Do Q13.1 through Q15.5.

Week 6:

- Master antiderivatives, substitution, and the Fundamental Theorem.
- Do Q16.1 through Q18.1.

Week 7:

- Master applications of integration.
- Do Q19.1 through Q19.9.
- Take Sample Final 1 and 2 under timed conditions.

How to use Part 5 while drilling: cover everything below the **Question** line, work the problem cold, then uncover the steps. Reading a worked solution is not practice; reproducing it is.

Part 9: Final checklist before exam day

- Calculator is set to radians.
- You know the derivative and integral tables cold.
- You know the special trig limits.
- You know the exact unit circle values.
- You can sketch from limit and derivative conditions.
- You can do a chain rule problem without dropping the inside derivative.
- You can write the quotient rule in the correct order.
- You add + C only to indefinite integrals.
- You split distance problems at the zeros of velocity.
- You use $b - a$ as the interval width when bounding or averaging.
- You convert degrees to radians before any linearization.
- You check whether a critical number comes from $f' = 0$ or from f' being undefined.
- You cancel common factors before declaring a vertical asymptote.
- You bracket whole numerators when typing into Möbius, and you click preview.
- You have timed yourself on a full sample paper.

The ten most expensive single mistakes in this course

Each of these is drawn from a specific question in Part 5.

1. $\sqrt{6} + \sqrt{2}/4$ instead of $(\sqrt{6} + \sqrt{2})/4$, which is off by a factor of nearly 3. See Q01.1.
2. Reporting a cancelled factor as a vertical asymptote instead of a hole. See Q04.1.
3. Using $dx = 2$ instead of $dx = \frac{\pi}{90}$ in a degree linearization. See Q12.2.
4. Reading "diameter equals height" as $r = h$ rather than $r = \frac{h}{2}$. See Q11.1.
5. Missing critical numbers where f' is undefined rather than zero. See Q15.5.
6. Integrating across a crossing point without splitting, which cancels area. See Q19.2.
7. Forgetting the chain-rule factor on a variable limit of integration. See Q18.1.
8. Rounding between Newton iterations. See Q13.1.
9. Assuming work scales linearly with spring stretch, when it scales with the square. See Q19.5.
10. Calling a horizontal tangent a maximum when f' does not change sign. See Q14.1b.